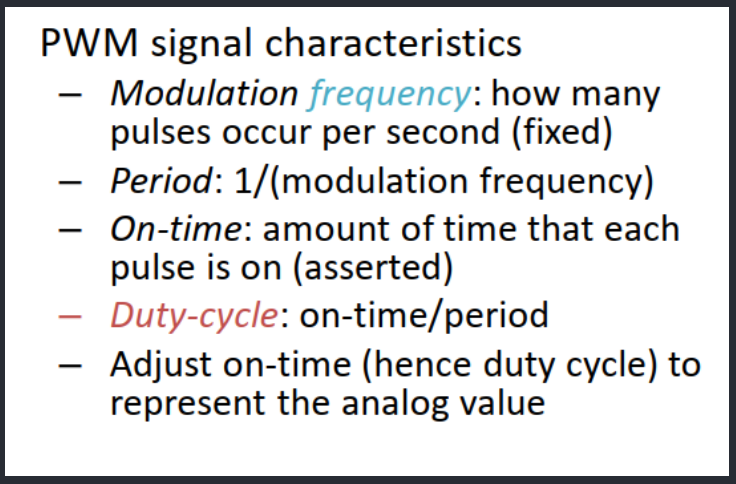
سوالات تحلیلی

سوال ۱: مشخصات یک سیگنال PWM چیست؟



(مدولاسیون) فرکانس: چند پالس بر ثانیه اتفاق می‌افتد

پریود/دوره: معکوس (مدولاسیون) فرکانس/دوره یک پالس

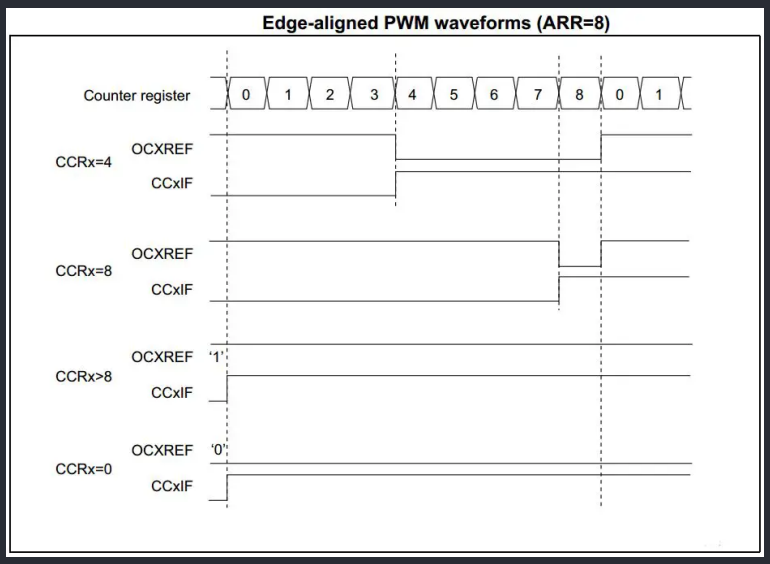
آن تایم/مدت زمان فعال بودن: مدت زمانی که پالس on/فعال است (صفر نیست)

دیوتی سایکل: درصد زمانی که پالس on/فعال است

سوال ۲: تفاوت مد یک و دو PWM در STM32F4 چیست؟

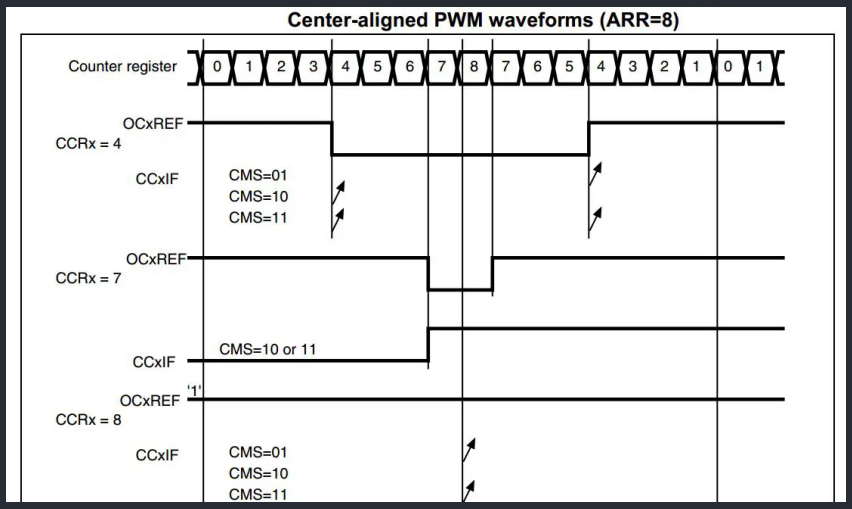
Mode1: Edge-Aligned Mode

این مود خود دارای دو کانفیگ Up-Counting و Down-Counting است که با رجیستر TIMx_CR1 قابل تنظیم است. با توجه به کانفیگ، سیگنال PWM ما (OCxREF) تا زمانی فعال است که به ترتیب TIMx_CCRx < TIMx_CNT و TIMx_CCRx < TIMx_CNT باشد.



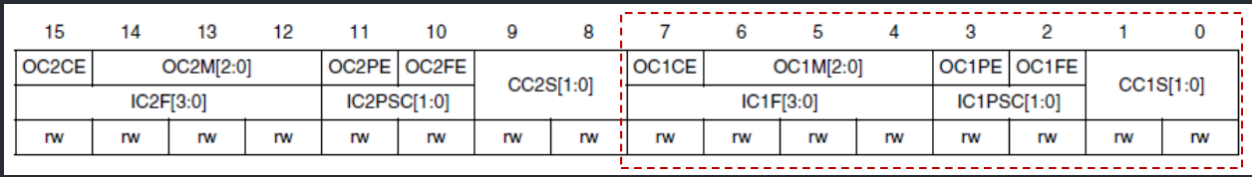
Mode2: Center-Aligned Mode

در این مود، compare flag زمانی ست می‌شود که counter ما یک بار پشت هم count-up و count-down (و یا بسته به کانفیگ آن جابجا) کند.



سوال ۳: رجیسترهای کانالهای capture / compare را نام ببرید و ذکر کنید در مدهای مختلف چه کاربردی دارند.

TIMx_CCMR1/2: Capture/Compare Mode Register



15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC2CE	OC2M[2:0]			OC2PE	OC2FE	CC2S[1:0]		OC1CE	OC1M[2:0]			OC1PE	OC1FE	CC1S[1:0]	
IC2F[3:0]				IC2PSC[1:0]				IC1F[3:0]				IC1PSC[1:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

- **CCxS** (Capture/Compare Selection): Input Capture mode
  - **CC1S[1:0]** in CCMR1:
    - 00: The channel is configured as input, mapped on TI1 (PA0 pin).
    - 01: The channel is configured as input, mapped on TI2 (PA1 pin).
    - 10: The channel is configured as input, mapped on TI3 (PA2 pin).
    - 11: The channel is configured as input, mapped on TI4 (PA3 pin).
  - **CC2S[1:0]** in CCMR1:
    - 00: The channel is configured as input, mapped on TI2 (PA1 pin).
    - 01: The channel is configured as input, mapped on TI1 (PA0 pin).
    - 10: The channel is configured as input, mapped on TI4 (PA3 pin).
    - 11: The channel is configured as input, mapped on TI3 (PA2 pin).
- **ICxPSC** (Input Capture Prescaler): determines the frequency at which input captures are performed.
- **ICxF** (Input Capture Filter)

- **CCxS** (Capture/Compare Selection): Output Compare mode
  - **CC1S[1:0]** CCMR1:
    - 00: The channel is configured as output, PWM mode 1.
    - 01: The channel is configured as output, PWM mode 2.
    - 10: The channel is configured as output, PWM mode 3.
    - 11: The channel is configured as output, PWM mode 4.
  - **CC2S[1:0]** CCMR1:
    - 00: The channel is configured as output, PWM mode 1.
    - 01: The channel is configured as output, PWM mode 2.
    - 10: The channel is configured as output, PWM mode 3.
    - 11: The channel is configured as output, PWM mode 4.
- **OCxFE** (Output Compare Fast Enable): In this mode, the output compare signal can change more quickly, but with a reduced resolution.

- **OCxPE** (Output Compare Preload Enable): When enabled, the compare value is loaded from a preload register, allowing for atomic updates and synchronization with the timer's counter.
- **OCxCE** (Output Compare Clear Enable)
- **OCxM[2:0]** (Output Compare Mode)
    - 000: Frozen
    - 001: Set on match
    - 010: Clear on match
    - 011: Toggle on match
    - 100: Force inactive
    - 101: Force active
    - 110: PWM mode 1: clear on overflow
    - 111: PWM mode 2: set on overflow

TIMx_CCER: Capture/Compare Enable Register

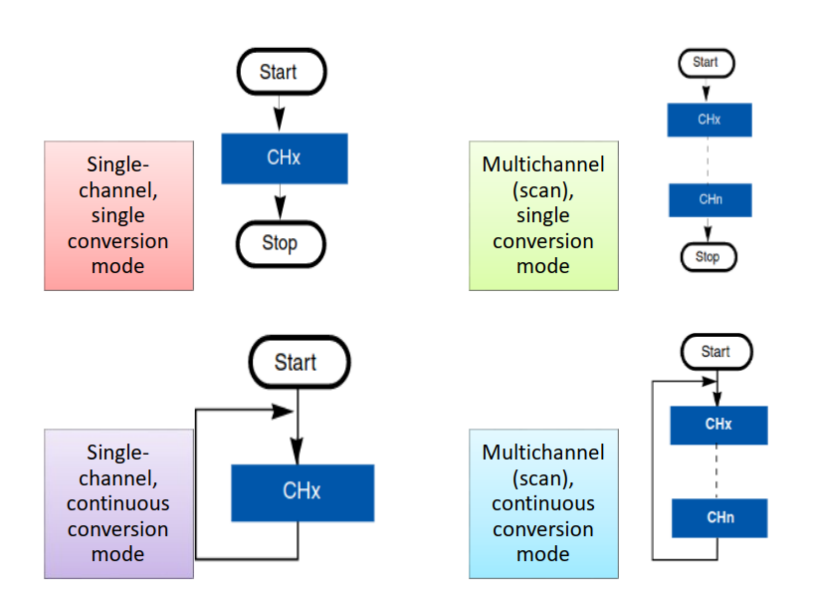
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CC4NP	Res.	CC4P	CC4E	CC3NP	Res.	CC3P	CC3E	CC2NP	Res.	CC2P	CC2E	CC1NP	Res.	CC1P	CC1E
rw		rw	rw	rw		rw	rw	rw		rw	rw	rw		rw	rw

- **CCxE**: enables the capture
- **CCxNP/CCxP**: determine to be sensetive to falling/rising edge

TIMx_CCR: Capture/Compare Register

سوال ۴: مُدهای کاری مختلف تبدیل آنالوگ به دیجیتال در میکروکنترلرهای STM32 را چیست و چه تفاوتی با هم دارند؟

دارای 4 مود است:



1) **Single-channel, single conversion mode:**

در این مود، کانورتر به طور همزمان تنها یک کانال ورودی آنالوگ را کانورت می‌کند و هر بار منتظر می‌ماند تا trigger شود و سپس کار کانورت کردن خود را آغاز کند.

2) **Multichannel (scan), single conversion mode:**

در این مود کانورتر به طور همزمان چندین (تا 16) کانال را کانورت می‌کند و مانند مود قبلی هر بار باید trigger شود.

3) **Single-channel continuous conversion mode:**

در این مود کانورتر تنها یک کانال را کانورت می‌کند، با این تفاوت که نیاز به trigger شدن ندارد و بلافاصله بعد از کانورت شدن سیگنال آنالوگ فعلی، به سراغ سیگنال بعدی می‌رود.

4) **Multichannel (scan) continuous conversion mode:**

در این مود کانورتر به طور همزمان چندین (تا 16) کانال را کانورت می‌کند و همانند مود قبلی نیاز به trigger شدن ندارد و پس از کانورت شدن سیگنال های فعلی، به سراغ سیگنال ها بعدی می‌رود.

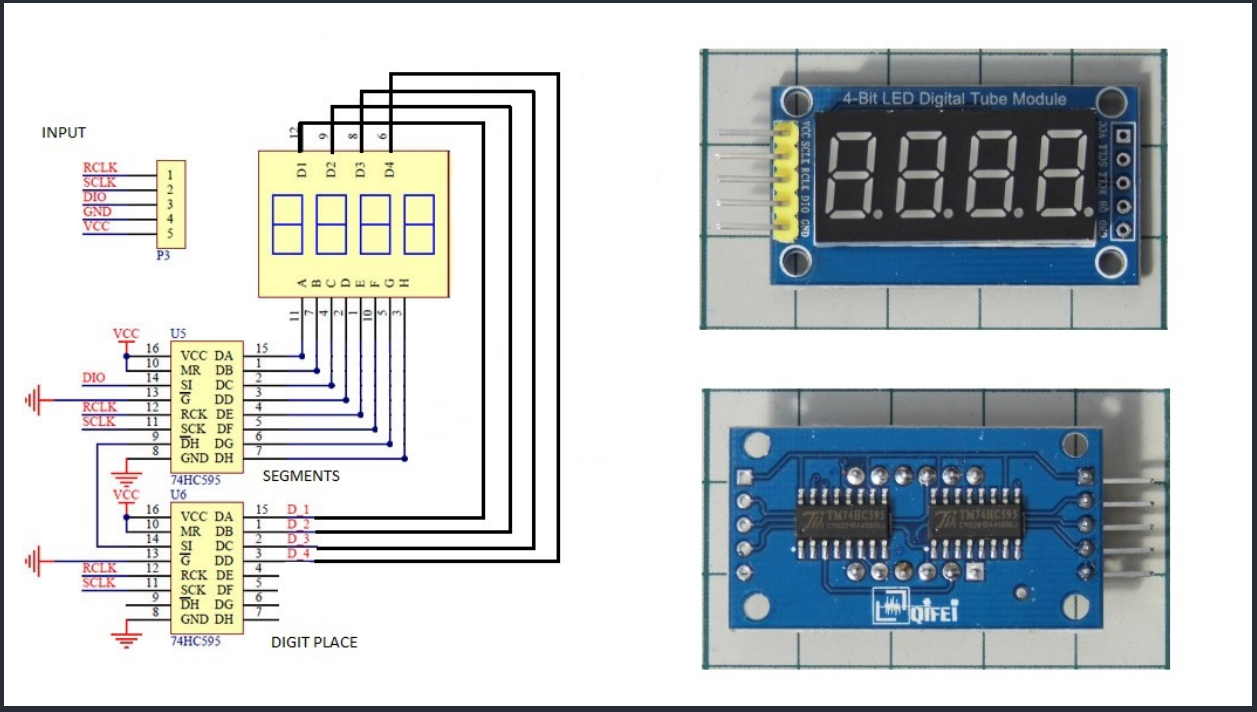
سوال ۵: تراشه 74595 چیست و چگونه کار میکند؟ یک digital tube چیست و چگونه با این تراشه ساخته میشود؟

تراشه 74HC595 یک IC شیفت رجیستر معروف است که کاربرد آن در داده های

- SIPO (Serial input Parallel output)

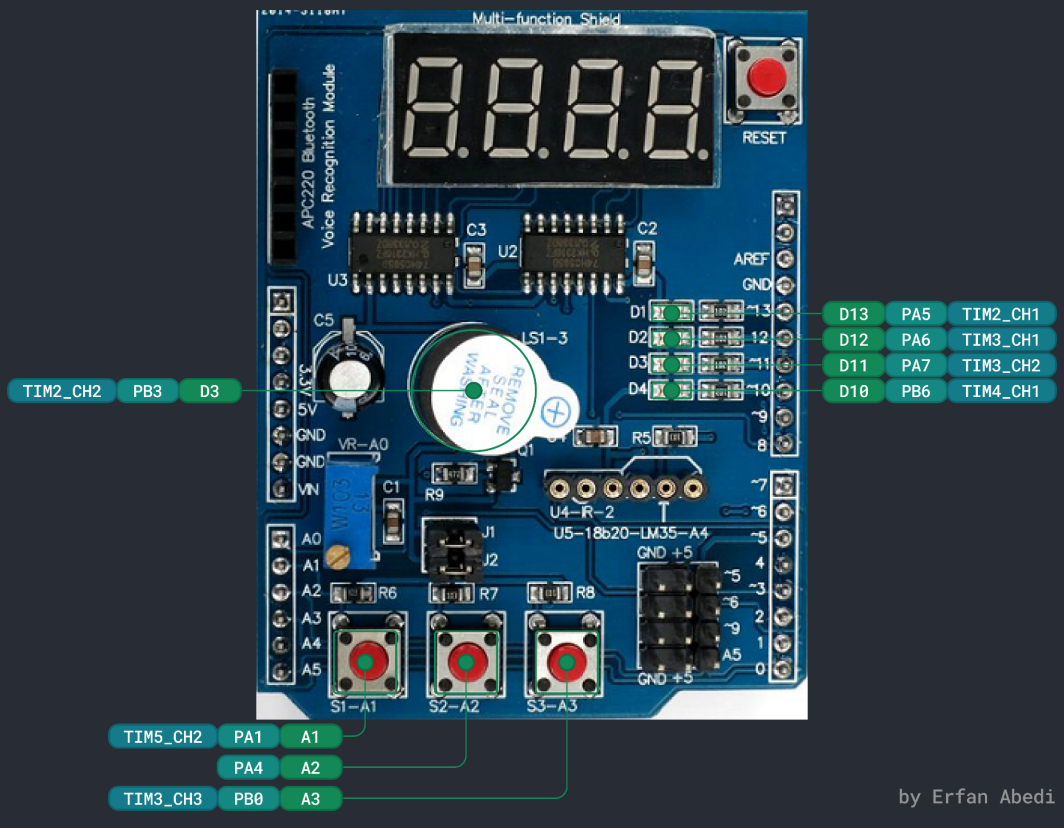
برای کنترل چندین خروجی با کاهش پین های اشغال شده ریزپردازنده ها یا مدارات است.

digital tubeها یا digital displayها صفحه نمایش هایی اند که برای نمایش چندین عدد یا کاراکتر استفاده میشوند و در ماشین حساب، ساعت های دیجیتالی و ادوات اندازه گیری الکترونیکی به کار میروند.



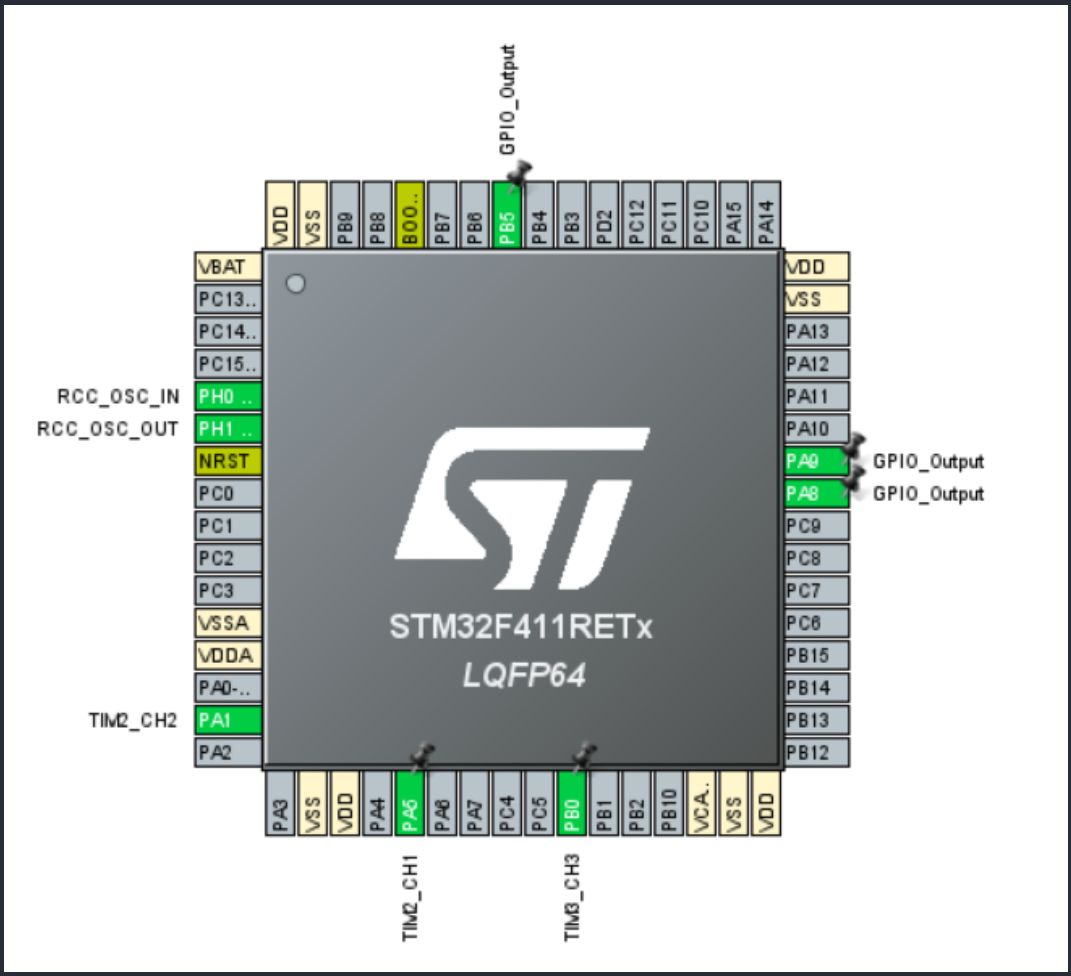
سوال ۶: در این آزمایشگاه از شیلد مولتی فانکشن آردوینو استفاده خواهد شد که ادوات متنوع و امکانات دیجیتال و آنالوگی روی خود دارد. با بررسی شماتیک و اتصالات شیلد، نقشه پینها (pinout) بورد Nucleo 64 مورد استفاده در آزمایشگاه و نیز جداول Alternate Function پین ها در بخش ۴ داده برگ (دیتاشیت) میکرو STM32F401 مشخص کنید هر یک از دکمه های فشاری، LEDها و بازر روی شیلد به کدام کانالهای کدام تایمرها میتوانند متصل شوند. پاسخ را در یک جدول ذکر کنید.

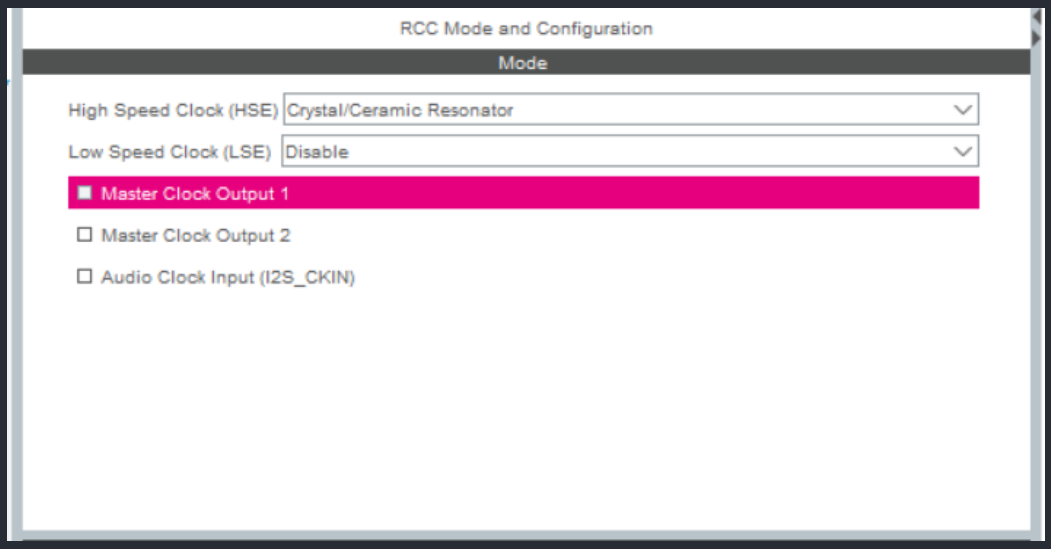
Components	Arduino Pins	STM32F401 Pins	AF01	AF02
LED1 / D1	D13	PA5	TIM2_CH1/ TIM2_ETR	-
LED2 / D2	D12	PA6	TIM1_BKIN	TIM3_CH1
LED3 / D3	D11	PA7	TIM1_CH1N	TIM3_CH2
LED4 / D4	D10	PB6	-	TIM4_CH1
Button1 / S1	A1	PA1	TIM2_CH2	TIM5_CH2
Button2 / S2	A2	PA4	-	-
Button3 / S3	A3	PB0	TIM1_CH2N	TIM3_CH3
Buzzer / LS1	D3	PB3	TIM2_CH2	-

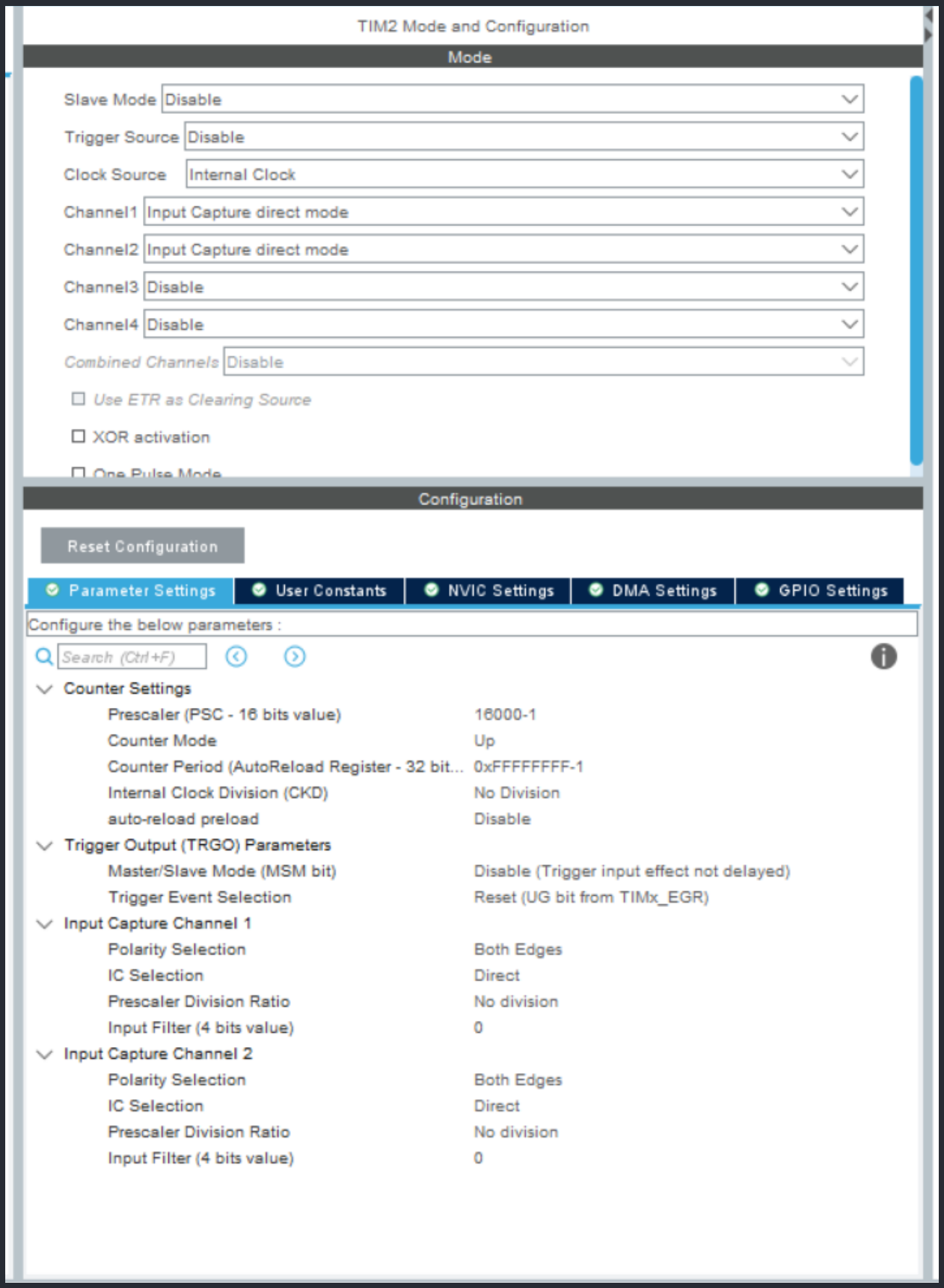


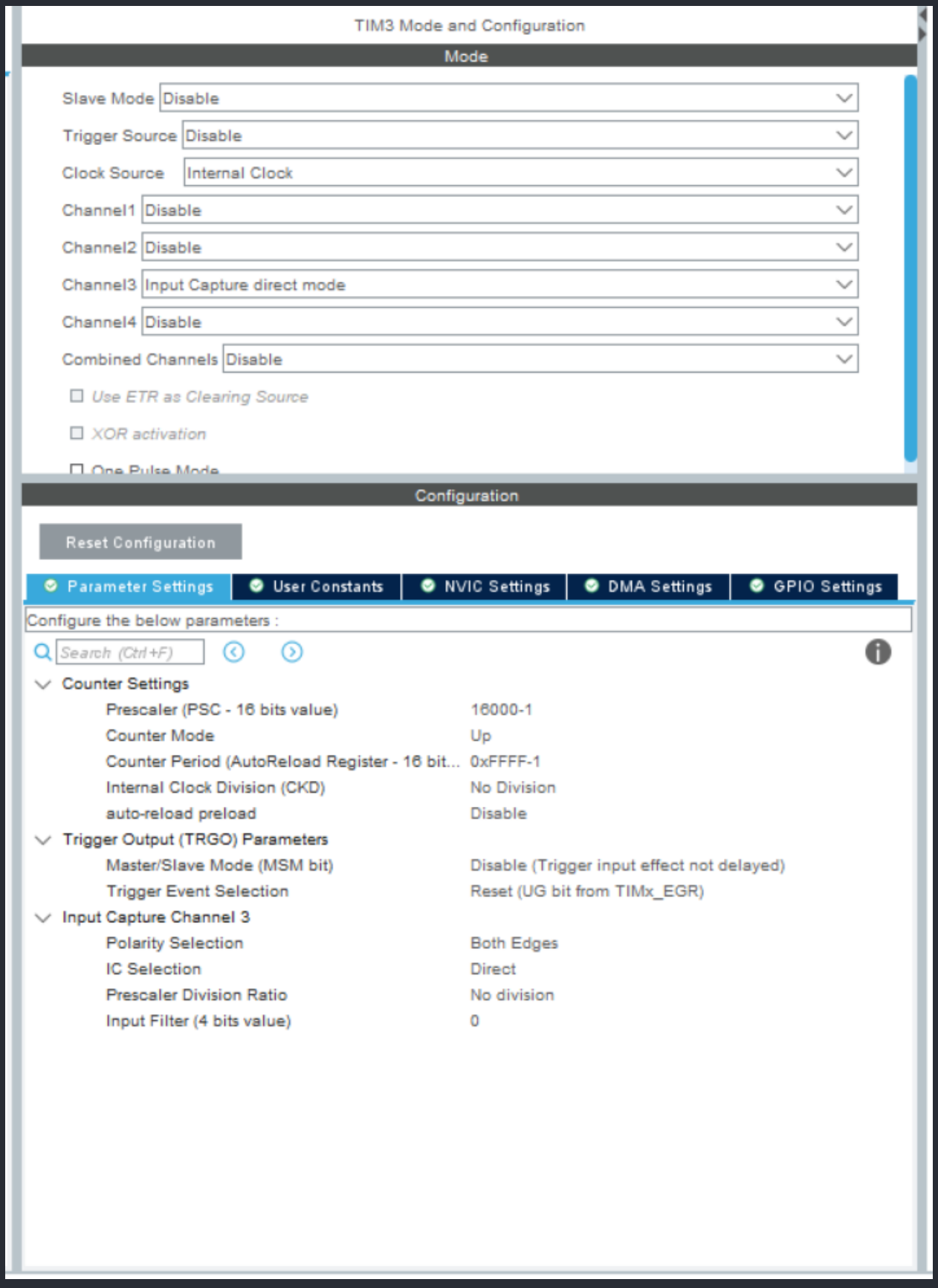
گزارش دستور کار 8:

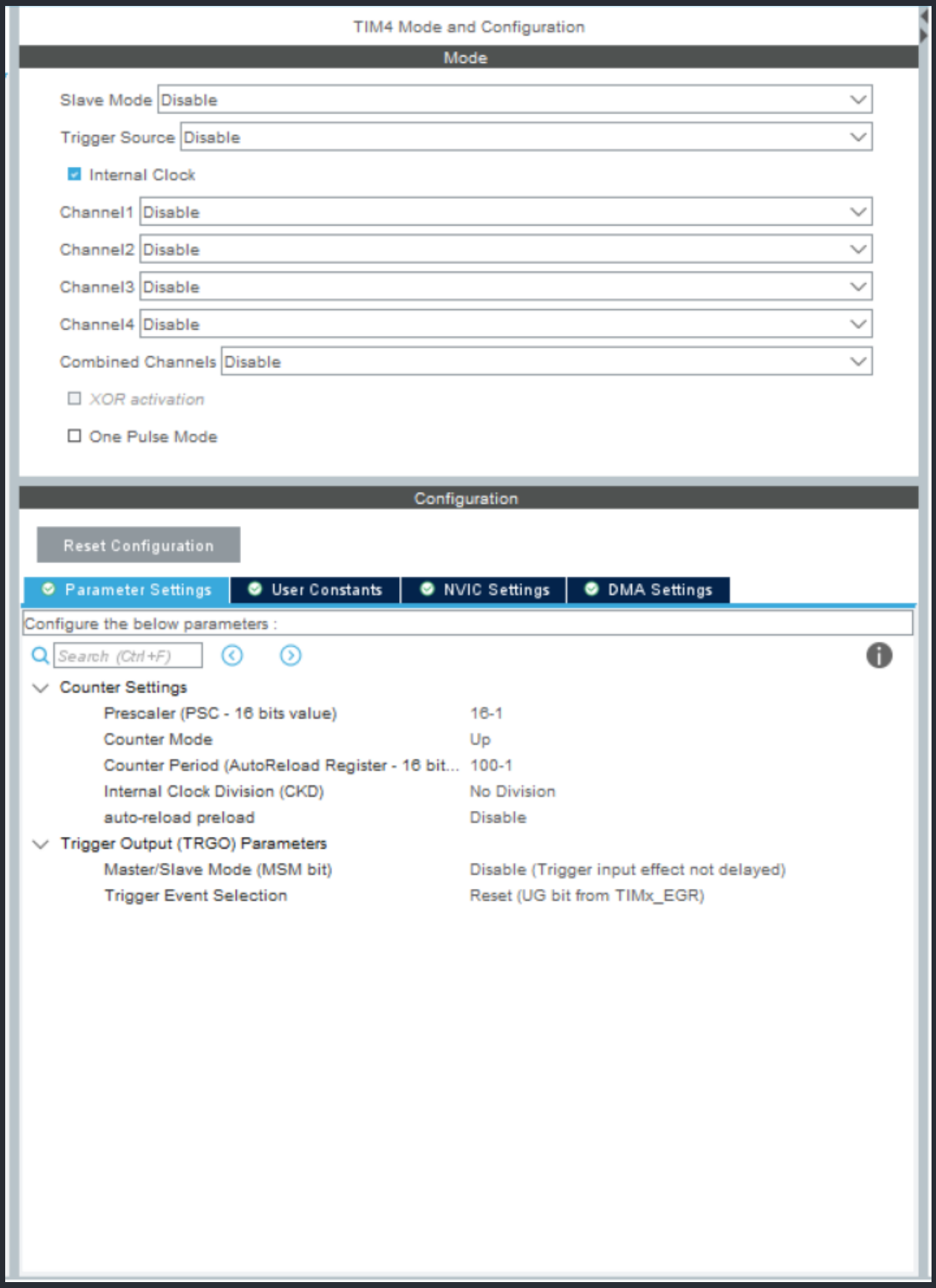
CubeMX:






TIM2 Mode and Configuration
Mode
Slave Mode Disable
Trigger Source Disable
Clock Source Internal Clock
Channel1 Input Capture direct mode
Channel2 Input Capture direct mode
Channel3 Disable
Channel4 Disable
Combined Channels Disable
Use ETR as Clearing Source
XOR activation
One Pulse Mode
Configuration
Reset Configuration
Parameter Settings
User Constants
NVIC Settings
DMA Settings
GPIO Settings
Configure the below parameters :
Search (Ctrl+F)
Counter Settings
Prescaler (PSC - 16 bits value) 16000-1
Counter Mode Up
Counter Period (AutoReload Register - 32 bit... 0xFFFFFFFF-1
Internal Clock Division (CKD) No Division
auto-reload preload Disable
Trigger Output (TRGO) Parameters
Master/Slave Mode (MSM bit) Disable (Trigger input effect not delayed)
Trigger Event Selection Reset (UG bit from TIMx_EGR)
Input Capture Channel 1
Polarity Selection Both Edges
IC Selection Direct
Prescaler Division Ratio No division
Input Filter (4 bits value) 0
Input Capture Channel 2
Polarity Selection Both Edges
IC Selection Direct
Prescaler Division Ratio No division
Input Filter (4 bits value) 0


TIM3 Mode and Configuration
Mode
Slave Mode Disable
Trigger Source Disable
Clock Source Internal Clock
Channel1 Disable
Channel2 Disable
Channel3 Input Capture direct mode
Channel4 Disable
Combined Channels Disable
Use ETR as Clearing Source
XOR activation
One Pulse Mode
Configuration
Reset Configuration
Parameter Settings
User Constants
NVIC Settings
DMA Settings
GPIO Settings
Configure the below parameters :
Search (Ctrl+F)
Counter Settings
Prescaler (PSC - 16 bits value) 16000-1
Counter Mode Up
Counter Period (AutoReload Register - 16 bit... 0xFFFF-1
Internal Clock Division (CKD) No Division
auto-reload preload Disable
Trigger Output (TRGO) Parameters
Master/Slave Mode (MSM bit) Disable (Trigger input effect not delayed)
Trigger Event Selection Reset (UG bit from TIMx_EGR)
Input Capture Channel 3
Polarity Selection Both Edges
IC Selection Direct
Prescaler Division Ratio No division
Input Filter (4 bits value) 0


TIM4 Mode and Configuration
Mode
Slave Mode
Disable
Trigger Source
Disable
Internal Clock
Channel1
Disable
Channel2
Disable
Channel3
Disable
Channel4
Disable
Combined Channels
Disable
XOR activation
One Pulse Mode
Configuration
Reset Configuration
Parameter Settings
User Constants
NVIC Settings
DMA Settings
Configure the below parameters :
Search (Ctrl+F)
Counter Settings
Prescaler (PSC - 16 bits value)
16-1
Counter Mode
Up
Counter Period (AutoReload Register - 16 bit...
100-1
Internal Clock Division (CKD)
No Division
auto-reload preload
Disable
Trigger Output (TRGO) Parameters
Master/Slave Mode (MSM bit)
Disable (Trigger input effect not delayed)
Trigger Event Selection
Reset (UG bit from TIMx_EGR)

کد:

main.c

```
40
41  /* USER CODE END PM */
42
43  /* Private variables ---------------------------------------------------------*/
44  TIM_HandleTypeDef htim2;
45  TIM_HandleTypeDef htim3;
46  TIM_HandleTypeDef htim4;
47
48  /* USER CODE BEGIN PV */
49  static const unsigned char sevenSegHex[10] = {0x3F, 0x06, 0x5B, 0x4F, 0x66,0x6D, 0x7D, 0x07, 0x7F, 0x6F};
50  static const unsigned char digitHex[4] = {0xFE, 0xFD, 0xFB, 0xF7};
51  volatile int N = 7;
52  volatile int M = 5;
53
54  int32_t IC_Val1 = 0;
55  int32_t IC_Val2 = 0;
56  uint32_t Difference = 0;
57  int Is_First_Captured = 0;
58  /* USER CODE END PV */
59
60  /* Private function prototypes -----------------------------------------------*/
61  void SystemClock_Config(void);
62  static void MX_GPIO_Init(void);
63  static void MX_TIM2_Init(void);
64  static void MX_TIM3_Init(void);
65  static void MX_TIM4_Init(void);
66  /* USER CODE BEGIN PFP */
67  int pascal(int row, int col);
68  void shiftOut(unsigned char num);
69  /* USER CODE END PFP */
70
```

main.c

```
72  /* USER CODE BEGIN 0 */
73  void HAL_TIM_IC_CaptureCallback(TIM_HandleTypeDef *htim)
74  {
75    if ((htim->Instance == TIM2 && htim->Channel == HAL_TIM_ACTIVE_CHANNEL_1) || (htim->Instance == TIM2 && htim->Channel == HAL_TIM_ACTIVE_CHANNEL_2))
76    {
77      if (Is_First_Captured == 0) // if the first rising edge is not captured
78      {
79        IC_Val1 = HAL_TIM_ReadCapturedValue(htim, ((htim->Channel == HAL_TIM_ACTIVE_CHANNEL_2) ? TIM_CHANNEL_2 : TIM_CHANNEL_1)); // read the first value
80        Is_First_Captured = 1;  // set the first captured as true
81      }
82
83      else   // If the first rising edge is captured, now we will capture the second edge
84      {
85        IC_Val2 = HAL_TIM_ReadCapturedValue(htim, ((htim->Channel == HAL_TIM_ACTIVE_CHANNEL_2) ? TIM_CHANNEL_2 : TIM_CHANNEL_1));  // read second value
86
87        if (IC_Val2 > IC_Val1)
88        {
89          Difference = IC_Val2 - IC_Val1;
90        }
91
92        else if (IC_Val1 > IC_Val2)
93        {
94          Difference = (((htim->Channel == HAL_TIM_ACTIVE_CHANNEL_2) ? PeriodTIM2 : PeriodTIM2) - IC_Val1) + IC_Val2;
95        }
96
97        if (htim->Channel == HAL_TIM_ACTIVE_CHANNEL_2)
98        {
99          N += ((Difference < 100) ? +1:-1);
100       }
101       else
102       {
103         M += ((Difference < 100) ? +1:-1);
104       }
105
106       __HAL_TIM_SET_COUNTER(htim, 0);  // reset the counter
107       Is_First_Captured = 0; // set it back to false
108     }
109   }
110 }
```

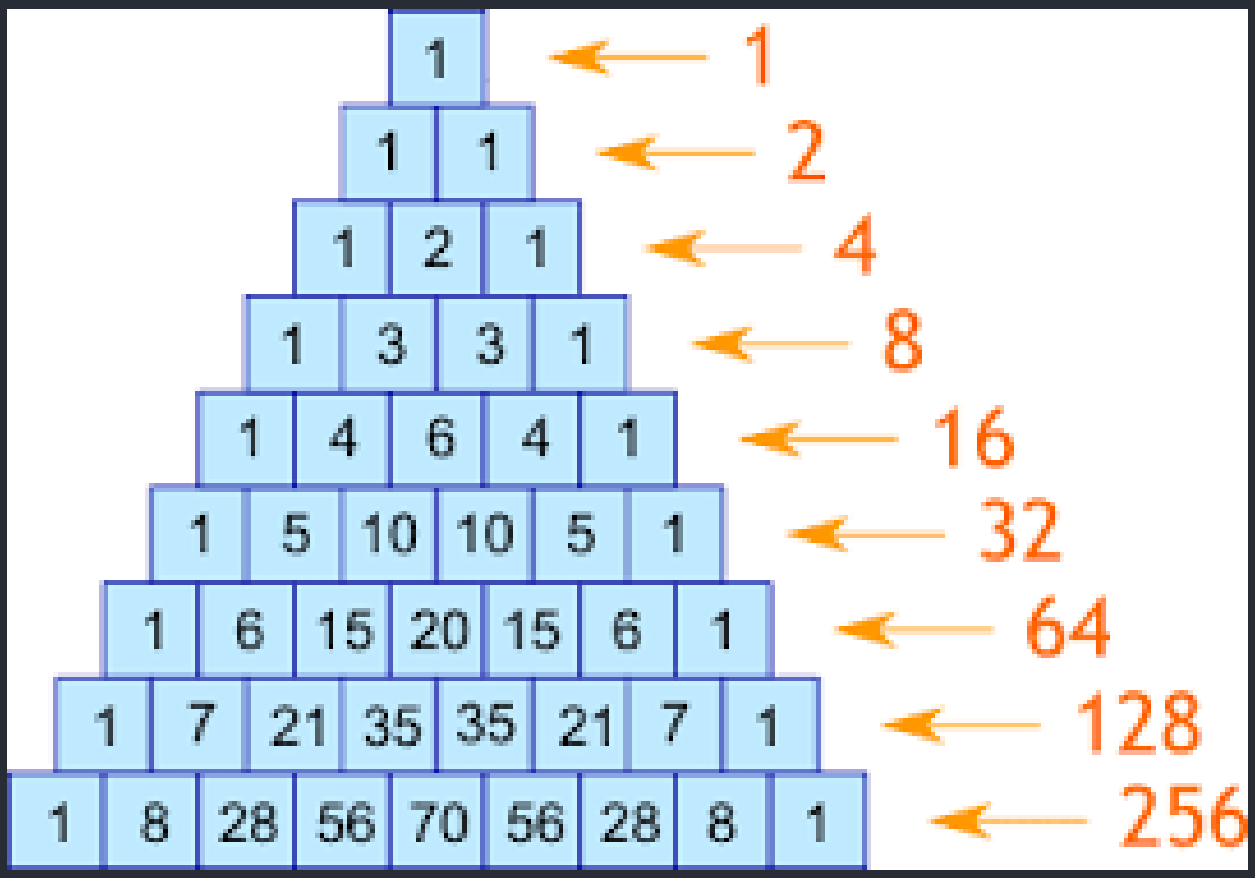
main.c

```
111  void HAL_TIM_PeriodElapsedCallback(TIM_HandleTypeDef *htim)
112  {
113    if (htim->Instance == TIM4)
114    {
115      int pasc = pascal(N, M);
116      int digits[4] = {(pasc / 1000) % 10, (pasc / 100) % 10, (pasc / 10) % 10, pasc % 10};
117
118      for(int i = 0; i < 4; i++)
119      {
120        HAL_GPIO_WritePin(GPIOB, GPIO_PIN_5, GPIO_PIN_RESET);
121        shiftOut(sevenSegHex[digits[i]]);
122        shiftOut(digitHex[i]);
123        HAL_GPIO_WritePin(GPIOB, GPIO_PIN_5, GPIO_PIN_SET);
124      }
125    }
126  }
127
128  void shiftOut(unsigned char num)
129  {
130    for(int i = 7; i >= 0; i--)
131    {
132      HAL_GPIO_WritePin(GPIOA, GPIO_PIN_8, GPIO_PIN_RESET);
133
134      if (((num >> i) & 0x01) == 0x01)
135        HAL_GPIO_WritePin(GPIOA, GPIO_PIN_9, GPIO_PIN_SET);
136      else
137        HAL_GPIO_WritePin(GPIOA, GPIO_PIN_9, GPIO_PIN_RESET);
138
139      HAL_GPIO_WritePin(GPIOA, GPIO_PIN_8, GPIO_PIN_SET);
140    }
141  }
142  int pascal(int row, int col)
143  {
144    if (col == 1 || row == col)
145      return 1;
146    else
147      return pascal(row - 1, col) + pascal(row - 1, col - 1);
148  }
149  /* USER CODE END 0 */
```

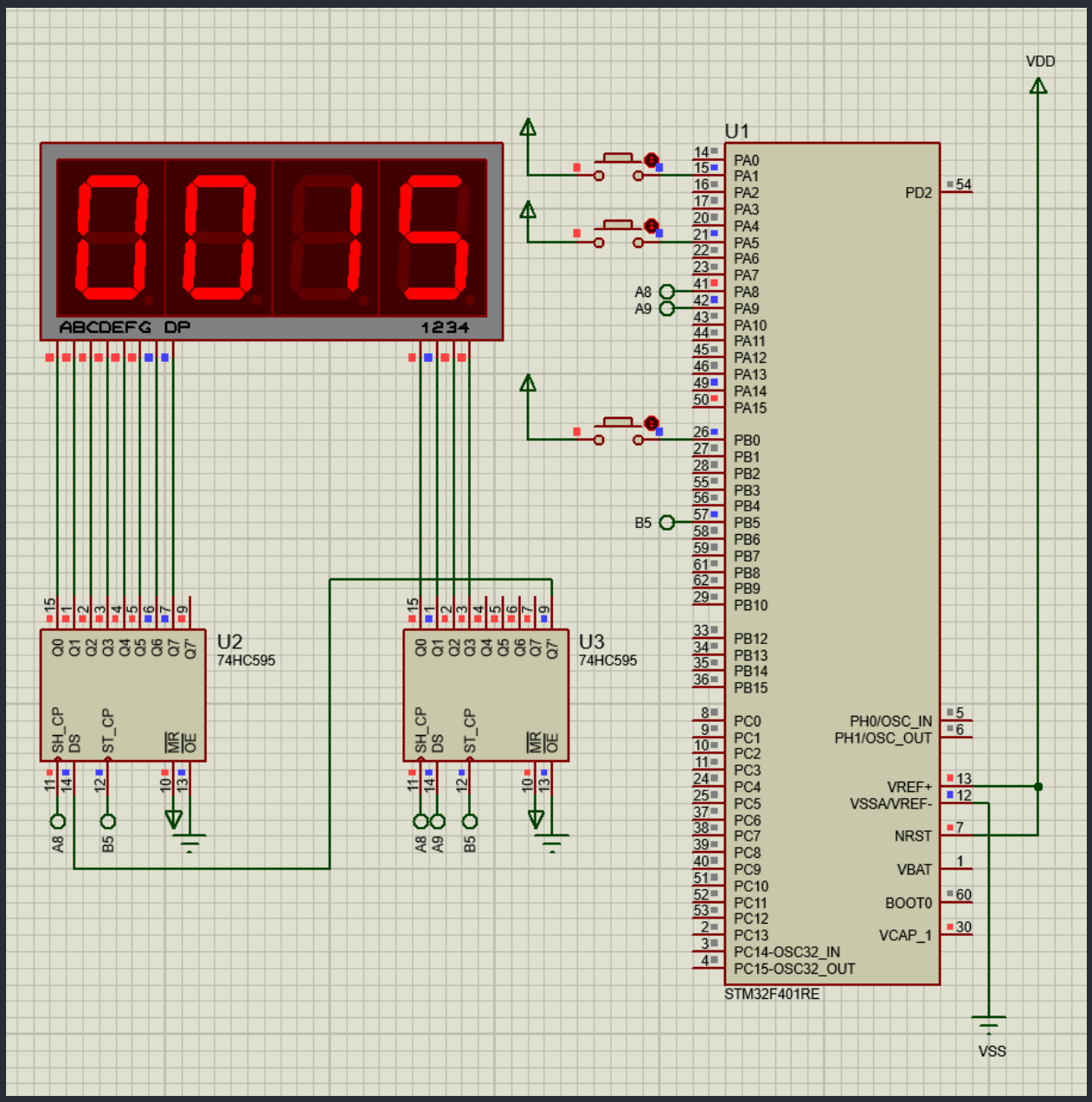
main.c

```
154    */
155   int main(void)
156  {
157    /* USER CODE BEGIN 1 */
158
159    /* USER CODE END 1 */
160
161    /* MCU Configuration--------------------------------------------------------*/
162
163    /* Reset of all peripherals, Initializes the Flash interface and the Systick. */
164    HAL_Init();
165
166    /* USER CODE BEGIN Init */
167
168    /* USER CODE END Init */
169
170    /* Configure the system clock */
171    SystemClock_Config();
172
173    /* USER CODE BEGIN SysInit */
174
175    /* USER CODE END SysInit */
176
177    /* Initialize all configured peripherals */
178    MX_GPIO_Init();
179    MX_TIM2_Init();
180    MX_TIM3_Init();
181    MX_TIM4_Init();
182    /* USER CODE BEGIN 2 */
183    HAL_TIM_Base_Start_IT(&htim4);
184    HAL_TIM_IC_Start_IT(&htim2, TIM_CHANNEL_1);
185    HAL_TIM_IC_Start_IT(&htim2, TIM_CHANNEL_2);
186    HAL_TIM_IC_Start_IT(&htim3, TIM_CHANNEL_3);
187    /* USER CODE END 2 */
188
189    /* Infinite loop */
190    /* USER CODE BEGIN WHILE */
191    while (1)
192    {
```

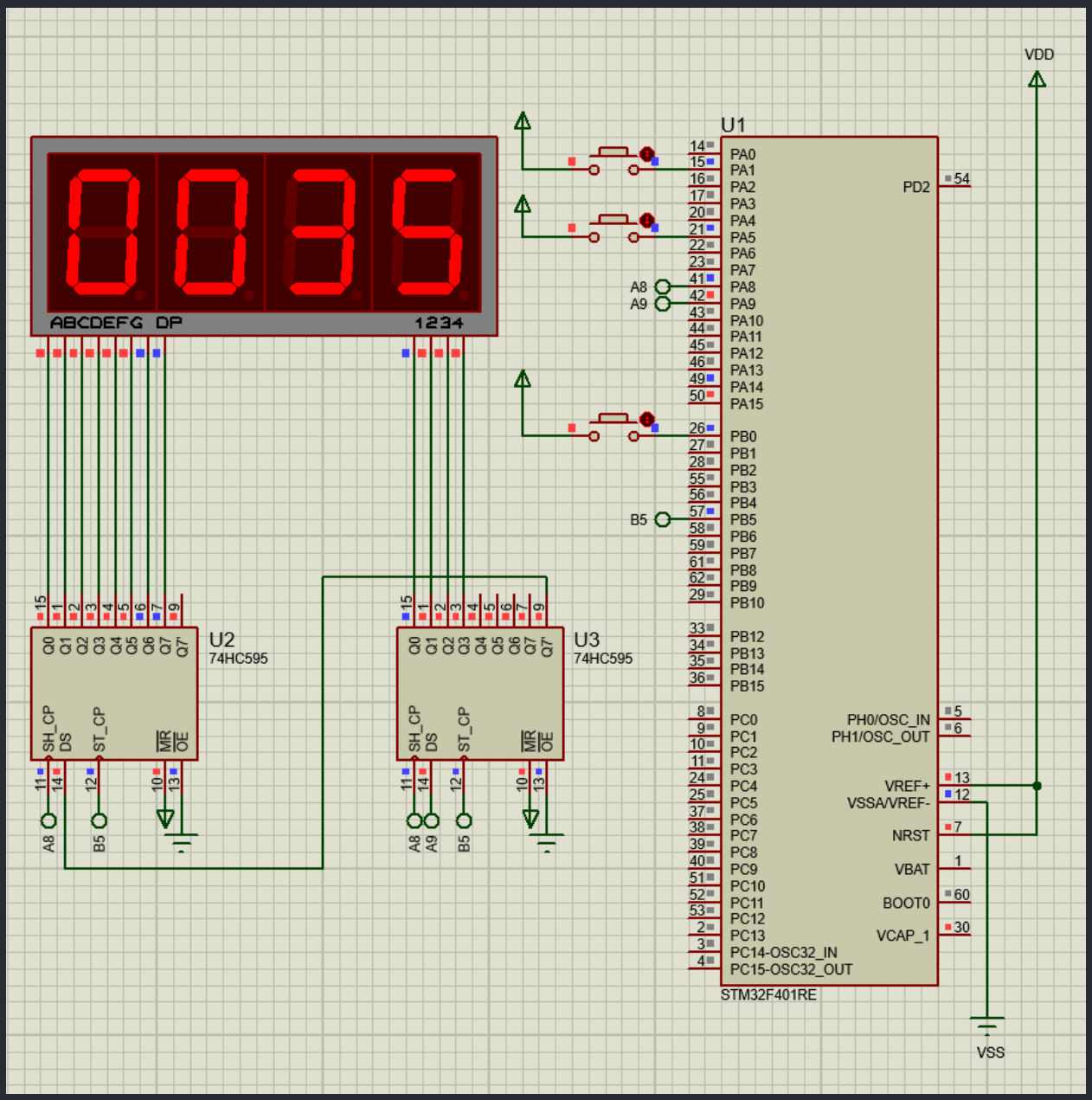
تست:


1 ← 1
1 1 ← 2
1 2 1 ← 4
1 3 3 1 ← 8
1 4 6 4 1 ← 16
1 5 10 10 5 1 ← 32
1 6 15 20 15 6 1 ← 64
1 7 21 35 35 21 7 1 ← 128
1 8 28 56 70 56 28 8 1 ← 256

N=7, M=5


VDD
U1
ABCDEFG DP
1234
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB12
PB13
PB14
PB15
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14-OSC32_IN
PC15-OSC32_OUT
PD2
PH0/OSC_IN
PH1/OSC_OUT
VREF+
VSSA/VREF-
NRST
VBAT
BOOT0
VCAP_1
STM32F401RE
A8
A9
B5
U2
74HC595
U3
74HC595
SH_CP
DS
ST_CP
MR
OE
VSS

N=8, M=5


VDD
U1
0035
ABCDEFG DP
1234
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
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PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
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PB9
PB10
PB12
PB13
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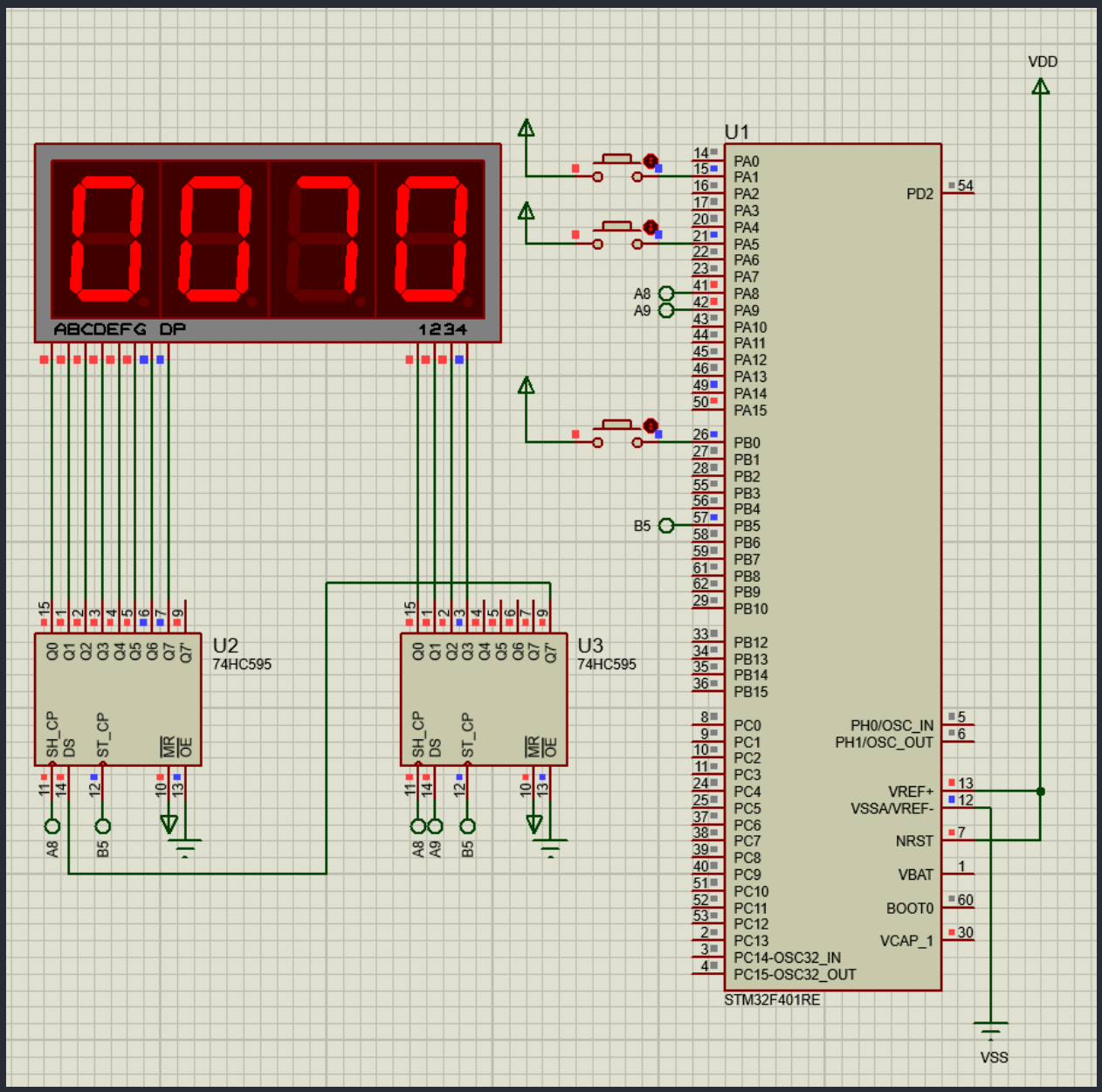
PB15
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14-OSC32_IN
PC15-OSC32_OUT
PD2
PH0/OSC_IN
PH1/OSC_OUT
VREF+
VSSA/VREF-
NRST
VBAT
BOOT0
VCAP_1
STM32F401RE
A8
A9
B5
U2
74HC595
U3
74HC595
Q0 Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q7'
SH_CP
DS
ST_CP
MR
OE
VSS

N=9, M=5


VDD
U1
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB12
PB13
PB14
PB15
PC0
PC1
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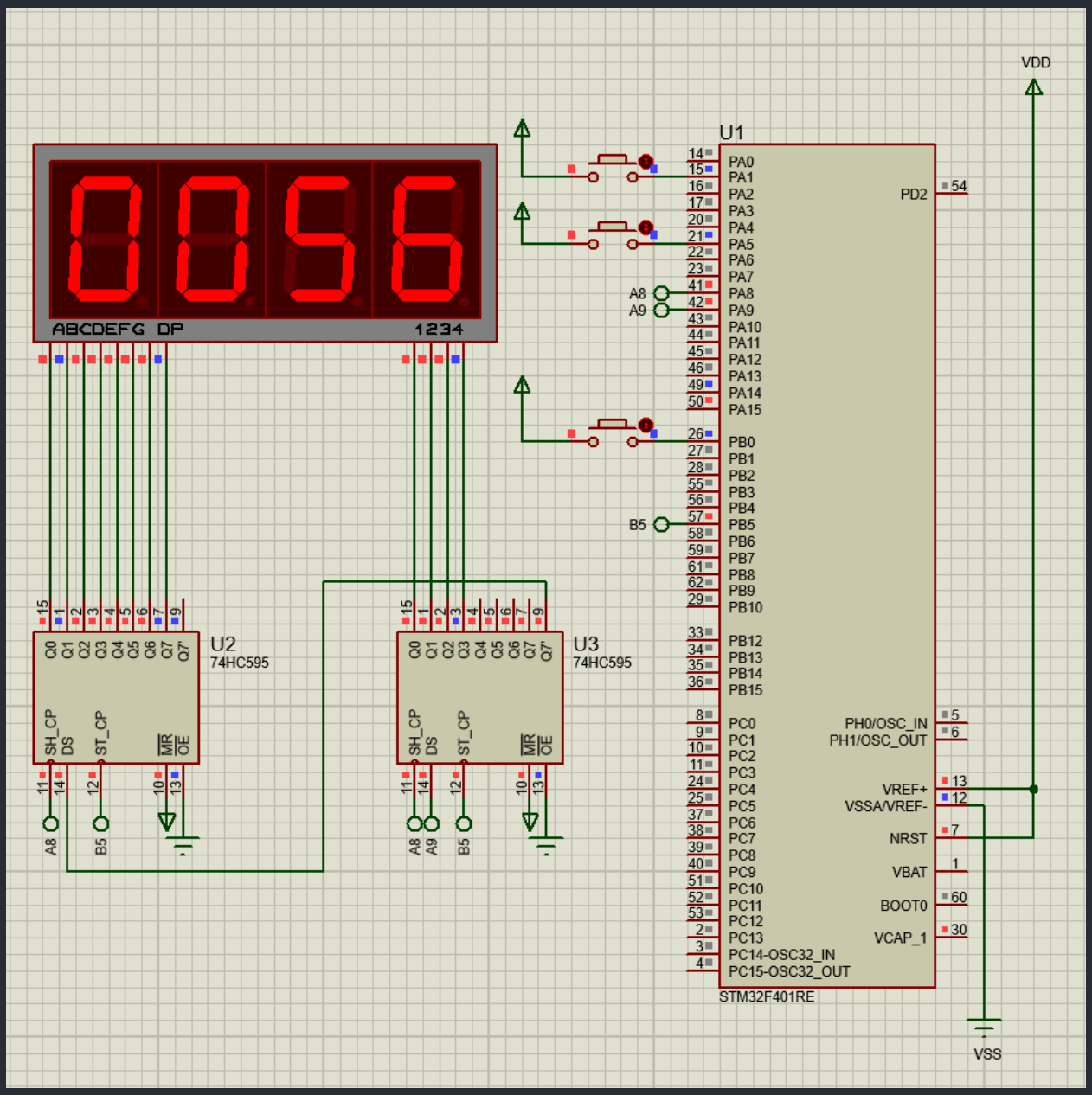
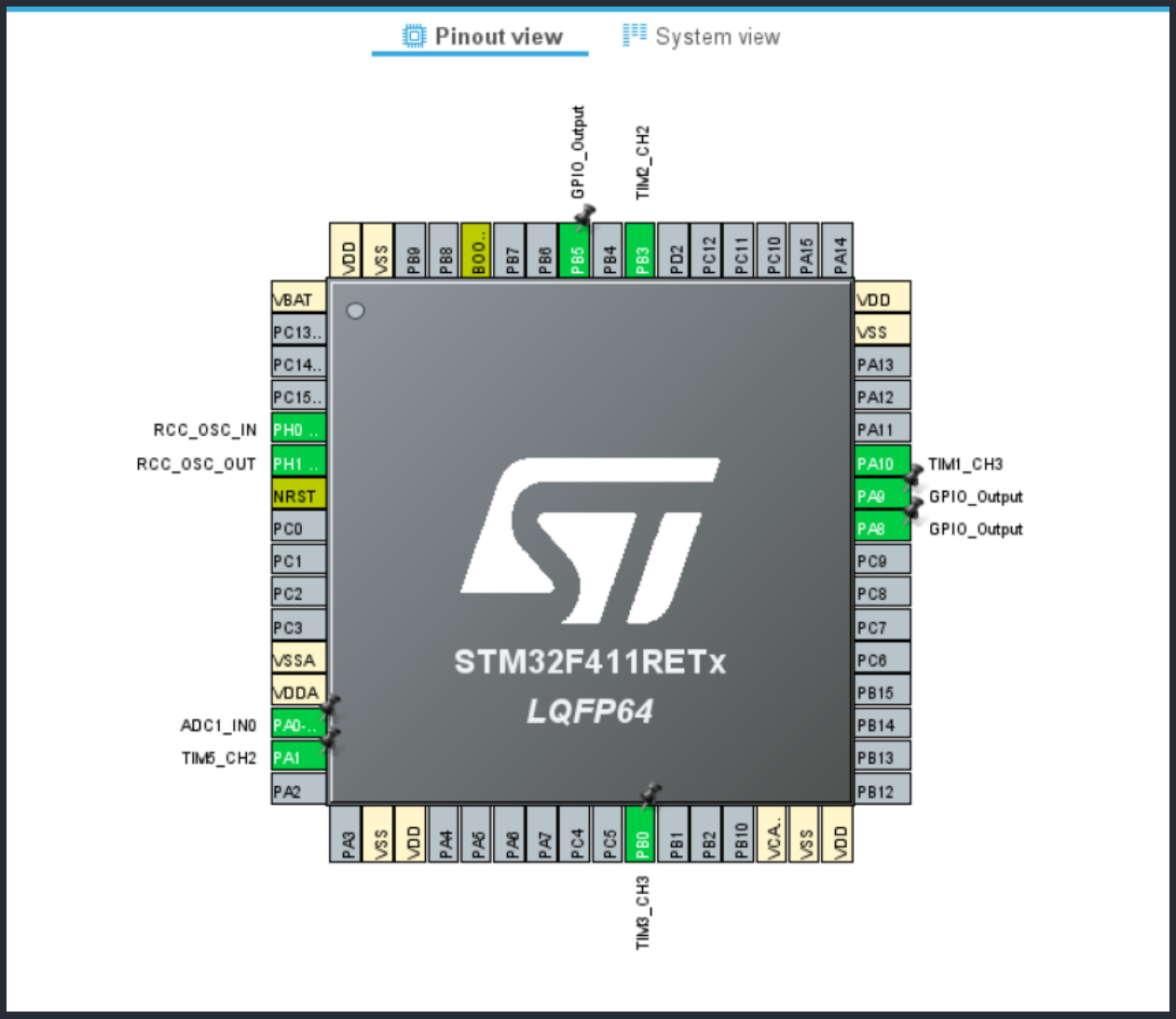
PC8
PC9
PC10
PC11
PC12
PC13
PC14-OSC32_IN
PC15-OSC32_OUT
PD2
PH0/OSC_IN
PH1/OSC_OUT
VREF+
VSSA/VREF-
NRST
VBAT
BOOT0
VCAP_1
STM32F401RE
A8
A9
B5
ABCDEFG DP
1234
U2
74HC595
U3
74HC595
SH_CP
DS
ST_CP
MR
OE
VSS

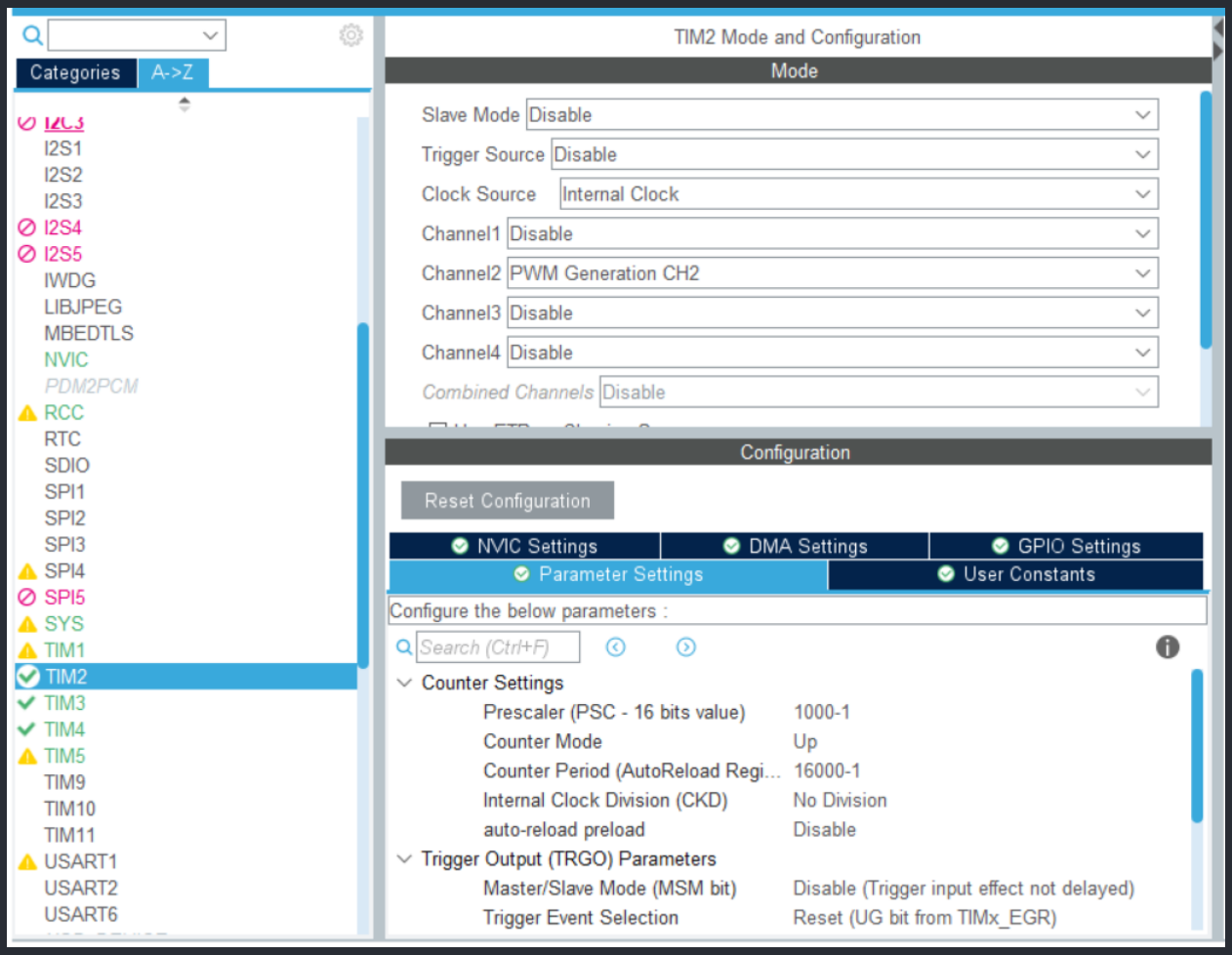
N=9, M=4

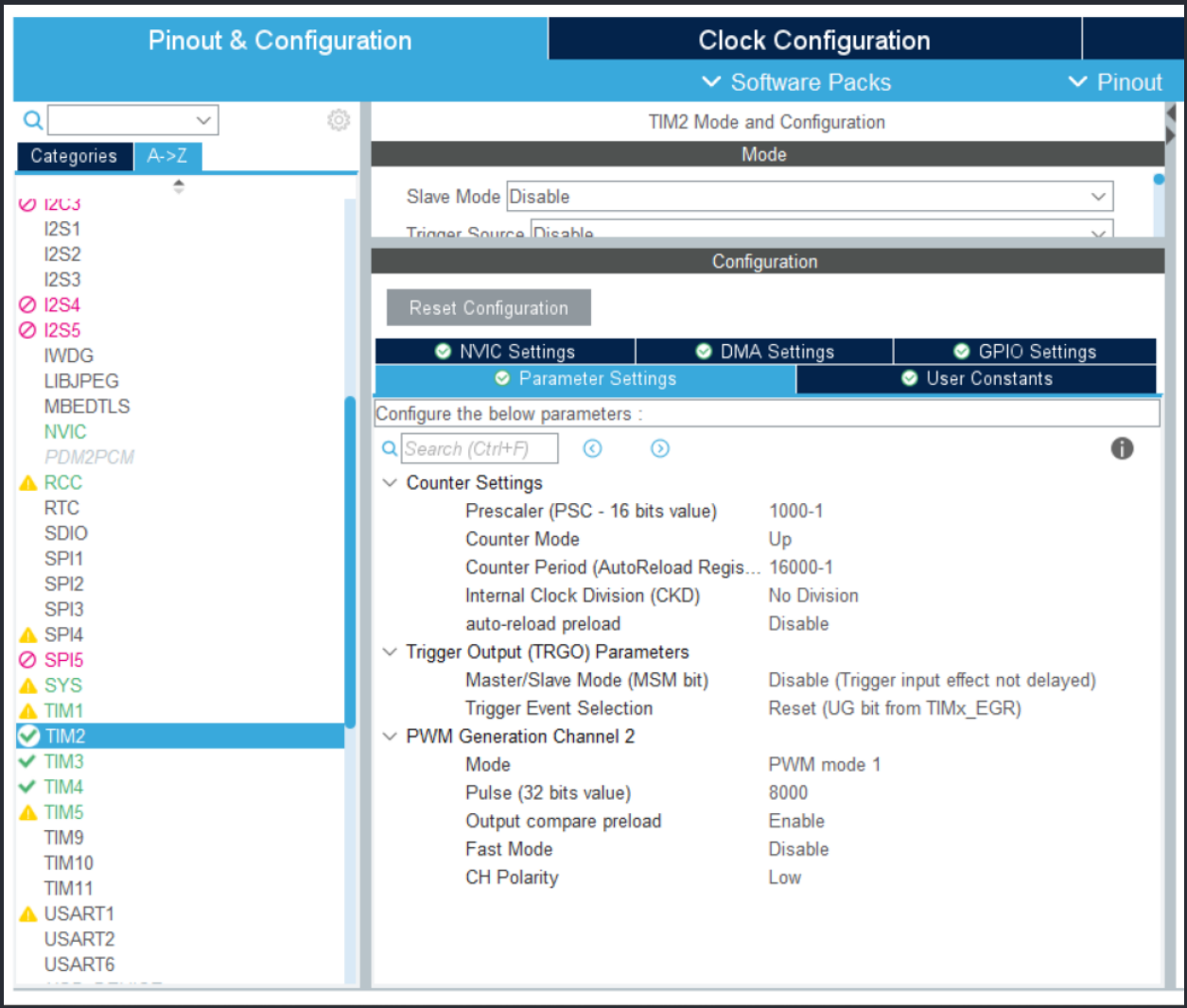

0056
ABCDEFG DP
1234
U1
STM32F401RE
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB12
PB13
PB14
PB15
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14-OSC32_IN
PC15-OSC32_OUT
PD2
PH0/OSC_IN
PH1/OSC_OUT
VREF+
VSSA/VREF-
NRST
VBAT
BOOT0
VCAP_1
VDD
VSS
A8
A9
B5
U2
74HC595
U3
74HC595
SH_CP
DS
ST_CP
MR
OE
Q0
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q7'

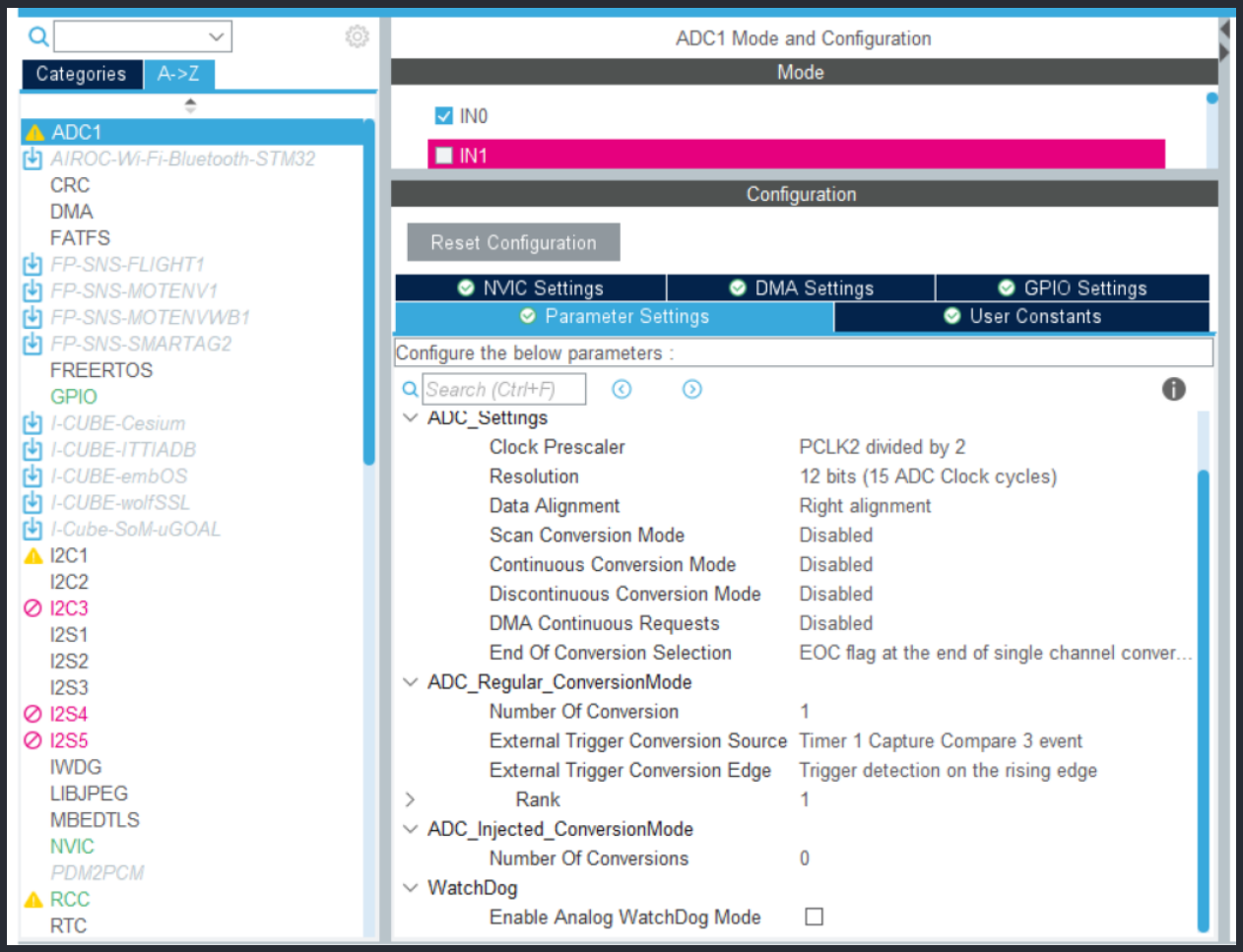
گزارش دستور کار 9:

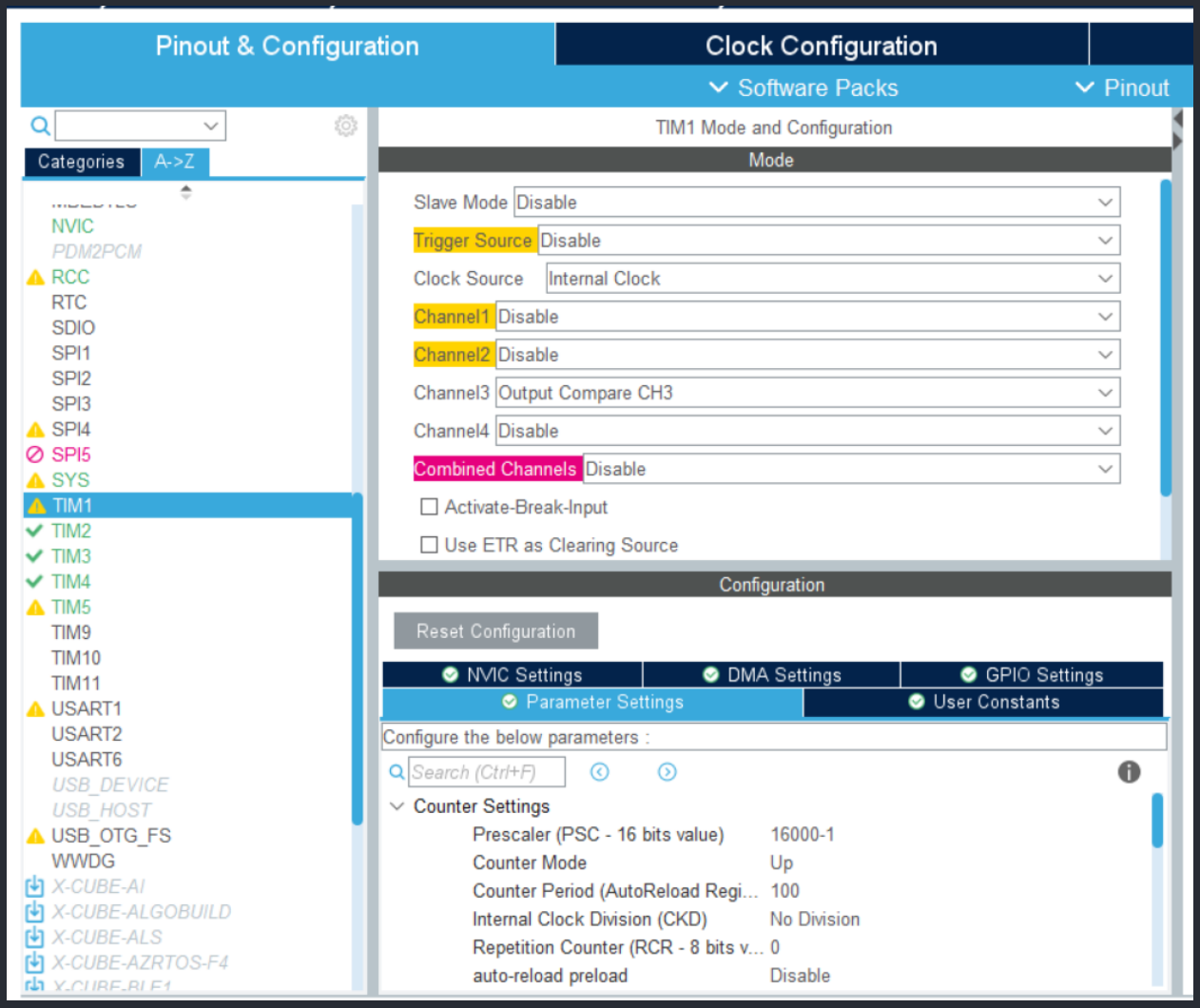
CubeMX:




Categories
A->Z
I2S1
I2S2
I2S3
I2S4
I2S5
IWDG
LIBJPEG
MBEDTLS
NVIC
PDM2PCM
RCC
RTC
SDIO
SPI1
SPI2
SPI3
SPI4
SPI5
SYS
TIM1
TIM2
TIM3
TIM4
TIM5
TIM9
TIM10
TIM11
USART1
USART2
USART6
TIM2 Mode and Configuration
Mode
Slave Mode Disable
Trigger Source Disable
Clock Source Internal Clock
Channel1 Disable
Channel2 PWM Generation CH2
Channel3 Disable
Channel4 Disable
Combined Channels Disable
Configuration
Reset Configuration
NVIC Settings
DMA Settings
GPIO Settings
Parameter Settings
User Constants
Configure the below parameters :
Search (Ctrl+F)
Counter Settings
Prescaler (PSC - 16 bits value) 1000-1
Counter Mode Up
Counter Period (AutoReload Regi... 16000-1
Internal Clock Division (CKD) No Division
auto-reload preload Disable
Trigger Output (TRGO) Parameters
Master/Slave Mode (MSM bit) Disable (Trigger input effect not delayed)
Trigger Event Selection Reset (UG bit from TIMx_EGR)


Pinout & Configuration
Clock Configuration
Software Packs
Pinout
Categories
A->Z
I2S1
I2S2
I2S3
I2S4
I2S5
IWDG
LIBJPEG
MBEDTLS
NVIC
PDM2PCM
RCC
RTC
SDIO
SPI1
SPI2
SPI3
SPI4
SPI5
SYS
TIM1
TIM2
TIM3
TIM4
TIM5
TIM9
TIM10
TIM11
USART1
USART2
USART6
TIM2 Mode and Configuration
Mode
Slave Mode
Disable
Configuration
Reset Configuration
NVIC Settings
DMA Settings
GPIO Settings
Parameter Settings
User Constants
Configure the below parameters :
Search (Ctrl+F)
Counter Settings
Prescaler (PSC - 16 bits value) 1000-1
Counter Mode Up
Counter Period (AutoReload Regis... 16000-1
Internal Clock Division (CKD) No Division
auto-reload preload Disable
Trigger Output (TRGO) Parameters
Master/Slave Mode (MSM bit) Disable (Trigger input effect not delayed)
Trigger Event Selection Reset (UG bit from TIMx_EGR)
PWM Generation Channel 2
Mode PWM mode 1
Pulse (32 bits value) 8000
Output compare preload Enable
Fast Mode Disable
CH Polarity Low


Categories
A->Z
ADC1
AIROC-Wi-Fi-Bluetooth-STM32
CRC
DMA
FATFS
FP-SNS-FLIGHT1
FP-SNS-MOTENV1
FP-SNS-MOTENVWB1
FP-SNS-SMARTAG2
FREERTOS
GPIO
I-CUBE-Cesium
I-CUBE-ITTIADB
I-CUBE-embOS
I-CUBE-wolfSSL
I-Cube-SoM-uGOAL
I2C1
I2C2
I2C3
I2S1
I2S2
I2S3
I2S4
I2S5
IWDG
LIBJPEG
MBEDTLS
NVIC
PDM2PCM
RCC
RTC
ADC1 Mode and Configuration
Mode
IN0
IN1
Configuration
Reset Configuration
NVIC Settings
DMA Settings
GPIO Settings
Parameter Settings
User Constants
Configure the below parameters :
Search (Ctrl+F)
ADC_Settings
Clock Prescaler
PCLK2 divided by 2
Resolution
12 bits (15 ADC Clock cycles)
Data Alignment
Right alignment
Scan Conversion Mode
Disabled
Continuous Conversion Mode
Disabled
Discontinuous Conversion Mode
Disabled
DMA Continuous Requests
Disabled
End Of Conversion Selection
EOC flag at the end of single channel conver...
ADC_Regular_ConversionMode
Number Of Conversion
1
External Trigger Conversion Source
Timer 1 Capture Compare 3 event
External Trigger Conversion Edge
Trigger detection on the rising edge
Rank
1
ADC_Injected_ConversionMode
Number Of Conversions
0
WatchDog
Enable Analog WatchDog Mode


Pinout & Configuration
Clock Configuration
Software Packs
Pinout
Categories
A->Z
NVIC
PDM2PCM
RCC
RTC
SDIO
SPI1
SPI2
SPI3
SPI4
SPI5
SYS
TIM1
TIM2
TIM3
TIM4
TIM5
TIM9
TIM10
TIM11
USART1
USART2
USART6
USB_DEVICE
USB_HOST
USB_OTG_FS
WWDG
X-CUBE-AI
X-CUBE-ALGOBUILD
X-CUBE-ALS
X-CUBE-AZRTOS-F4
TIM1 Mode and Configuration
Mode
Slave Mode Disable
Trigger Source Disable
Clock Source Internal Clock
Channel1 Disable
Channel2 Disable
Channel3 Output Compare CH3
Channel4 Disable
Combined Channels Disable
Activate-Break-Input
Use ETR as Clearing Source
Configuration
Reset Configuration
NVIC Settings
DMA Settings
GPIO Settings
Parameter Settings
User Constants
Configure the below parameters :
Search (Ctrl+F)
Counter Settings
Prescaler (PSC - 16 bits value) 16000-1
Counter Mode Up
Counter Period (AutoReload Regi... 100
Internal Clock Division (CKD) No Division
Repetition Counter (RCR - 8 bits v... 0
auto-reload preload Disable

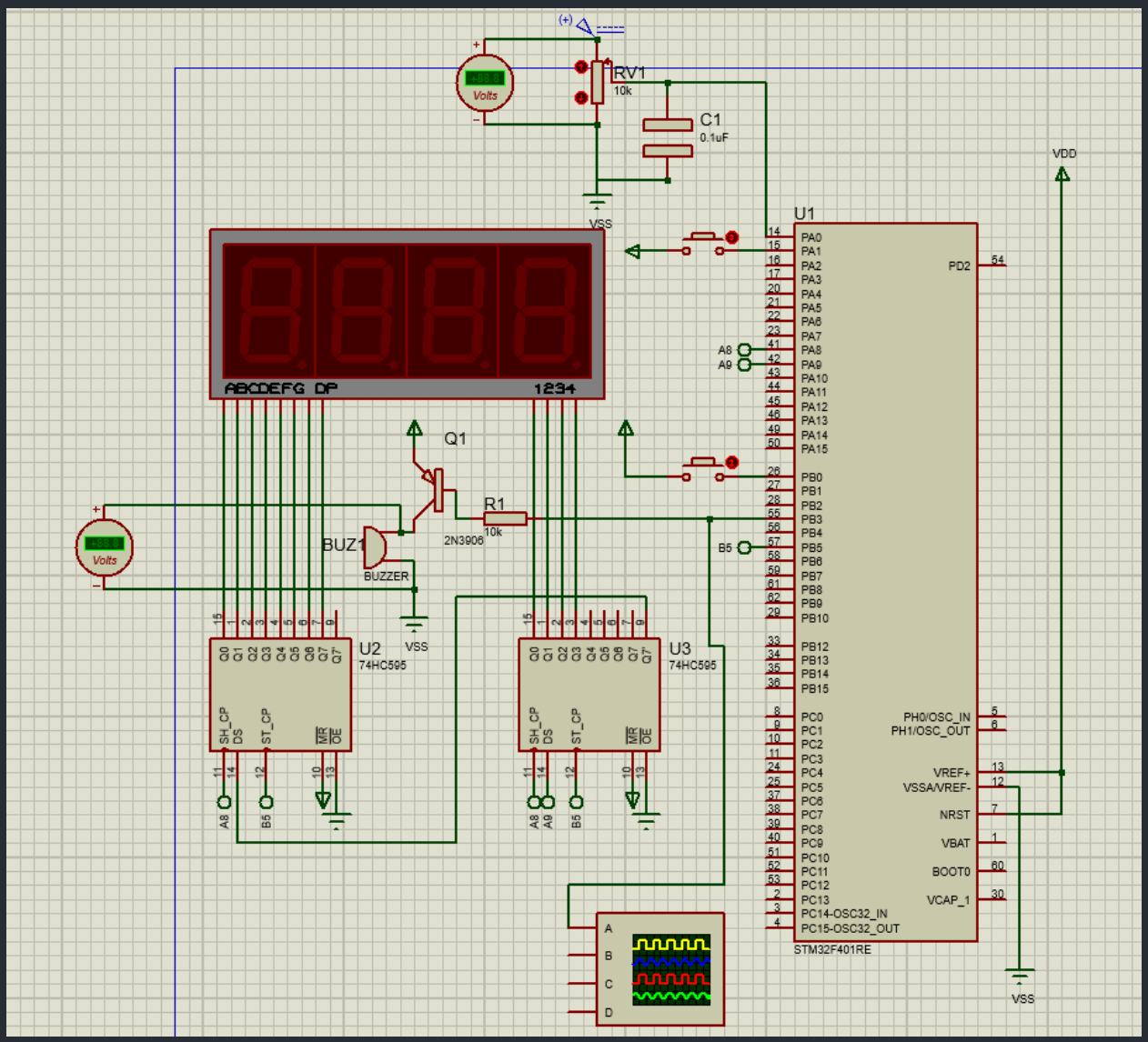
Code:

```
107       if (htim->Channel == HAL_TIM_ACTIVE_CHANNEL_2)
108       {
109         if (!(N == 1 && (Difference >= threshold)))
110           N += ((Difference < threshold) ? +1:-1);
111
112           htim2.Instance->PSC = adc_value/2;
113           HAL_TIM_PWM_Start(&htim2, TIM_CHANNEL_2);
114           HAL_Delay((Difference < threshold) ? 200:400);
115           HAL_TIM_PWM_Stop(&htim2, TIM_CHANNEL_2);
116       }
117       else
118       {
119         if (!((M == N && (Difference < threshold)) || (M == 1 && (Difference >= threshold))))
120           M += ((Difference < threshold) ? +1:-1);
121
122           htim2.Instance->PSC = adc_value;
123           HAL_TIM_PWM_Start(&htim2, TIM_CHANNEL_2);
124           HAL_Delay((Difference < threshold) ? 200:400);
125           HAL_TIM_PWM_Stop(&htim2, TIM_CHANNEL_2);
126       }
127
128       __HAL_TIM_SET_COUNTER(htim, 0);  // reset the counter
129       Is_First_Captured = 0; // set it back to false
130     }
131   }
132 }
```

```
180 void HAL_ADC_ConvCpltCallback(ADC_HandleTypeDef *hadc)
181 {
182   adc_value = HAL_ADC_GetValue(hadc);
183   HAL_ADC_Start_IT(&hadc1);
184 }
185
186
187 // Read the ADC value
188 uint32_t readADC(void)
189 {
190 // Start the ADC conversion
191 HAL_ADC_Start(&hadc1);
192
193 // Wait for the conversion to complete
194 HAL_ADC_PollForConversion(&hadc1, 1000);
195
196 // Get the converted value
197 uint32_t value = HAL_ADC_GetValue(&hadc1);
198
199 // Return the value
200 return value;
201 }
202
```

```
209  int main(void)
210  {
211    /* USER CODE BEGIN 1 */
212
213    /* USER CODE END 1 */
214
215    /* MCU Configuration--------------------------------------------------------*/
216
217    /* Reset of all peripherals, Initializes the Flash interface and the Systick. */
218    HAL_Init();
219
220    /* USER CODE BEGIN Init */
221
222    /* USER CODE END Init */
223
224    /* Configure the system clock */
225    SystemClock_Config();
226
227    /* USER CODE BEGIN SysInit */
228
229    /* USER CODE END SysInit */
230
231    /* Initialize all configured peripherals */
232    MX_GPIO_Init();
233    MX_TIM3_Init();
234    MX_TIM4_Init();
235    MX_TIM5_Init();
236    MX_TIM2_Init();
237    MX_ADC1_Init();
238    MX_TIM1_Init();
239    /* USER CODE BEGIN 2 */
240    HAL_TIM_Base_Start_IT(&htim4);
241    HAL_TIM_IC_Start_IT(&htim5, TIM_CHANNEL_2);
242    HAL_TIM_IC_Start_IT(&htim3, TIM_CHANNEL_3);
243    HAL_ADC_Start_IT(&hadc1);
244    /* USER CODE END 2 */
245
246    /* Infinite loop */
247    /* USER CODE BEGIN WHILE */
248    while (1)
```

Proteus Schema:



بخش امتیازی:

این تکه کد ها در این بخش اضافه شده اند:

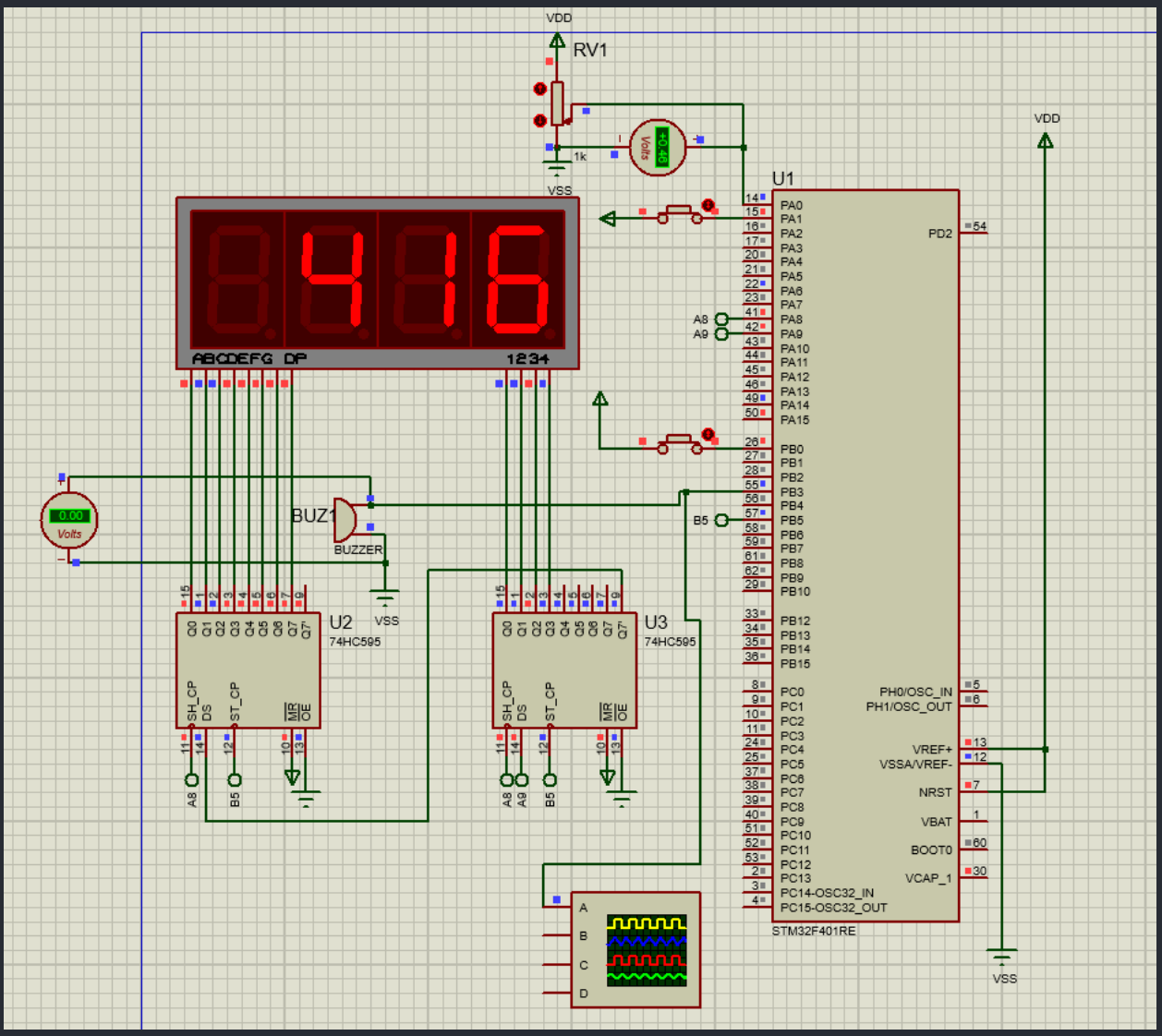
```
volatile int pressed_buttons_cnt = 0; // counter
volatile int two_buttons_pressed = 0; // flag
```

```
/* Private user code ---------------------------------------------------------*/
/* USER CODE BEGIN 0 */
void HAL_TIM_IC_CaptureCallback(TIM_HandleTypeDef *htim)
{
  if ((htim->Instance == TIM3 && htim->Channel == HAL_TIM_ACTIVE_CHANNEL_3) || (htim->Instance == TIM5 && htim->Channel == HAL_TIM_ACTIVE_CHANNEL_2))
  {
    if (Is_First_Captured == 0) // if the first rising edge is not captured
    {
      IC_Val1 = HAL_TIM_ReadCapturedValue(htim, ((htim->Channel == HAL_TIM_ACTIVE_CHANNEL_2) ? TIM_CHANNEL_2 : TIM_CHANNEL_3)); // read the first value
      Is_First_Captured = 1;  // set the first captured as true
      pressed_buttons_cnt++;
      if(pressed_buttons_cnt == 2) {
        two_buttons_pressed = 1;
      }
    }

    else if(two_buttons_pressed == 0) // If the first rising edge is captured, now we will capture the second edge
    {
      IC_Val2 = HAL_TIM_ReadCapturedValue(htim, ((htim->Channel == HAL_TIM_ACTIVE_CHANNEL_2) ? TIM_CHANNEL_2 : TIM_CHANNEL_3));  // read second value

      if (IC_Val2 > IC_Val1)
      {
        Difference = IC_Val2 - IC_Val1;
      }

      else if (IC_Val1 > IC_Val2)
      {
        Difference = (((htim->Channel == HAL_TIM_ACTIVE_CHANNEL_2) ? PeriodTIM5 : PeriodTIM3) - IC_Val1) + IC_Val2;
      }

      if (htim->Channel == HAL_TIM_ACTIVE_CHANNEL_2)
      {
        if (!(N == 1 && (Difference >= threshold)))
          N += ((Difference < threshold) ? +1:-1);

        if(adc_value != -1) {
          htim2.Instance->PSC = adc_value/2;
          HAL_TIM_PWM_Start(&htim2, TIM_CHANNEL_2);
          adc_delay_mode = (Difference < threshold) ? 1:20;
          buzzer_flag = 1;
```

```
127             buzzer_flag = 1;
128           }
129         }
130         else
131         {
132           if (!((M == N && (Difference < threshold)) || (M == 1 && (Difference >= threshold))))
133             M += ((Difference < threshold) ? +1:-1);
134
135             htim2.Instance->PSC = adc_value;
136             HAL_TIM_PWM_Start(&htim2, TIM_CHANNEL_2);
137             adc_delay_mode = (Difference < threshold) ? 1:2;
138             buzzer_flag = 1;
139         }
140
141         __HAL_TIM_SET_COUNTER(htim, 0);  // reset the counter
142         Is_First_Captured = 0; // set it back to false
143       }
144       else
145       {
146         pressed_buttons_cnt--;
147         if(pressed_buttons_cnt == 0) {
148           two_buttons_pressed = 0; // set flag FALSE
149         }
150         __HAL_TIM_SET_COUNTER(htim, 0);  // reset the counter
151         Is_First_Captured = 0;
152       }
153     }
154   }
```

```
157  void HAL_TIM_PeriodElapsedCallback(TIM_HandleTypeDef *htim)
158  {
159    if (htim->Instance == TIM4)
160    {
161      if (buzzer_flag == 1) {
162        ctr +=1;
163        if (ctr == adc_delay_mode * 2e5) // TODO: 2e5
164        {
165          HAL_TIM_PWM_Stop(&htim2, TIM_CHANNEL_2);
166          ctr = 0;
167        }
168      }
169
170      adc_ctr += 1;
171      if(adc_ctr == 1e3) { // TODO: 1e5
172        adc_value = readADC();
173        htim2.Instance->PSC = adc_value;
174        adc_ctr = 0;
175      }
176
177      if(two_buttons_pressed == 1) {
178        int buzzer_freq = cal_buzzer_freq();
179        lcd_show_number((buzzer_freq < 1 ? 0:buzzer_freq)%10000);
180      }
181      else {
182        lcd_show_number(pascal(N, M));
183      }
184    }
185  }
```

نمونه خروجی:


VDD
RV1
1k
VSS
U1
STM32F401RE
U2
74HC595
U3
74HC595
BUZ1
BUZZER
ABCDEFG DP
1234
0.00
Volts
A8
A9
B5
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
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PB13
PB14
PB15
PC0
PC1
PC2
PC3
PC4
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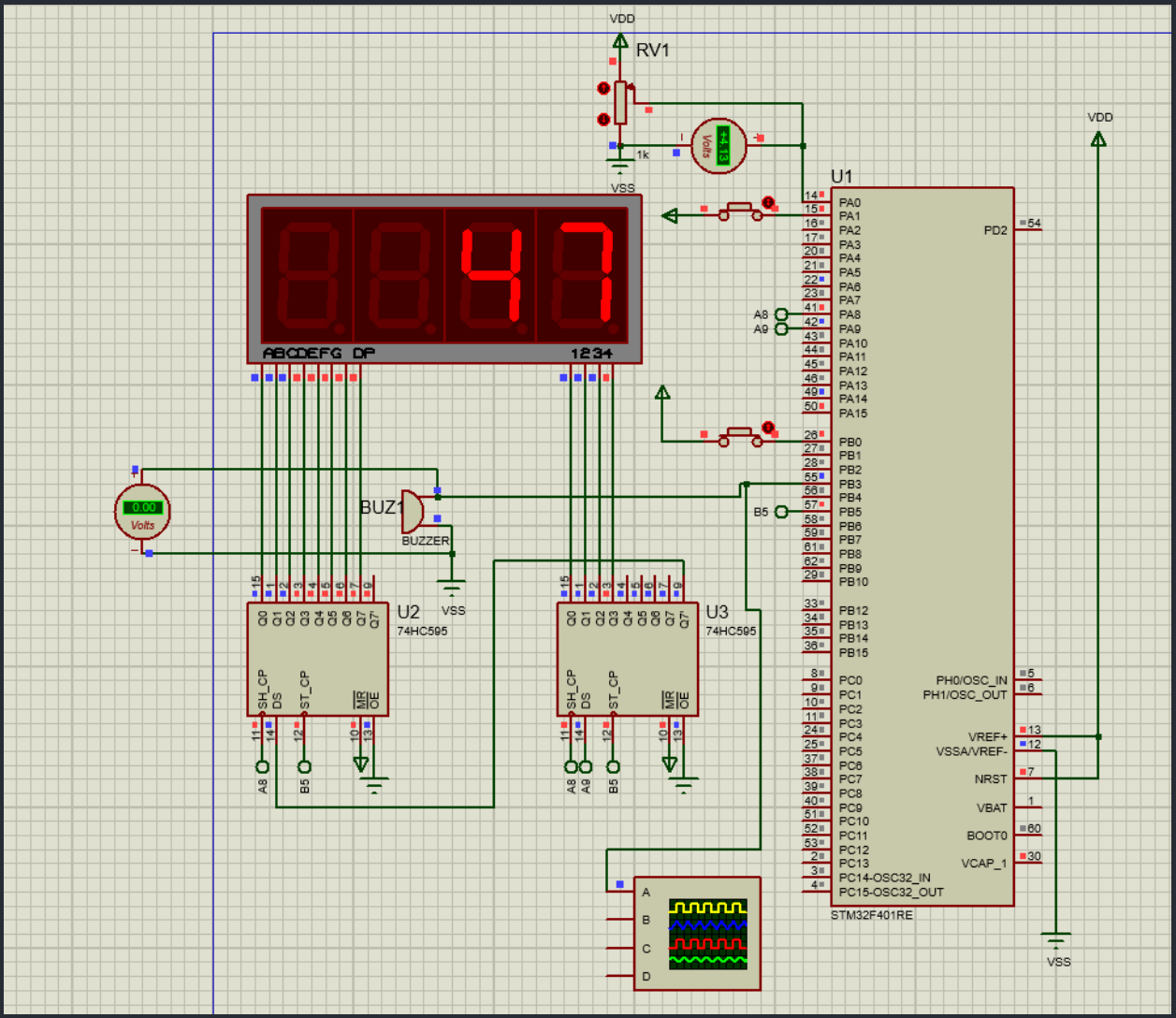
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14-OSC32_IN
PC15-OSC32_OUT
PD2
PH0/OSC_IN
PH1/OSC_OUT
VREF+
VSSA/VREF-
NRST
VBAT
BOOT0
VCAP_1
SH_CP
DS
ST_CP
MR
OE
A
B
C
D


VDD
RV1
1k
VSS
U1
STM32F401RE
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
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PA9
PA10
PA11
PA12
PA13
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PA15
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PC0
PC1
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PC8
PC9
PC10
PC11
PC12
PC13
PC14-OSC32_IN
PC15-OSC32_OUT
PD2
PH0/OSC_IN
PH1/OSC_OUT
VREF+
VSSA/VREF-
NRST
VBAT
BOOT0
VCAP_1
A8
A9
B5
BUZ1
BUZZER
U2
74HC595
U3
74HC595
SH_CP
DS
ST_CP
MR
OE
ABCDEFG DP
1234
A
B
C
D